

SOME APPLICATIONS OF THE MODULI SPACE OF $U \oplus D_8(-1)$ -POLARIZED K3 SURFACES

ADRIAN CLINGER, ANDREAS MALMENDIER, AND BRANDON WILLIAMS

ABSTRACT. We study K3 surfaces polarized by the lattice $U \oplus D_8(-1)$ through elliptic fibrations and orthogonal modular forms. Using an alternate genus-one fibration and its relative Jacobian, we construct an explicit map from Vinberg's coefficient space to the Hashimoto–Ueda family. We also explain the geometric meaning of certain character forms associated with these moduli spaces and extend the relative-Jacobian construction to the exceptional lattices $U \oplus E_8(-1) \oplus A_k(-1)$ with $k = 1, 2$.

1. INTRODUCTION

It has been noted that, in certain cases of K3 surfaces, Weierstrass form coefficients associated to distinguished elliptic fibrations often carry automorphic properties. A classical example is given by the Igusa generators attached to the Jacobians of smooth genus-two curves and their associated Kummer surfaces. For their Shioda–Inose partners, namely the family of $U \oplus E_8(-1) \oplus E_7(-1)$ -polarized K3 surfaces, the even generators occur as Weierstrass coefficients, and the odd generator χ_{35} is the modular Jacobian of the even generators [26, 33]. In particular, the cusp form χ_{10} cuts out the Humbert component $\mathcal{H}_1$, whereas the odd generator χ_{35} cuts out the full reflective ramification divisor $\mathcal{H}_1 \cup \mathcal{H}_4$. Its Borcherds product is also the denominator of the corresponding automorphic correction [34].

In [10], the authors obtained an elliptic normal form for the family of $U \oplus D_8(-1)$ -polarized K3 surfaces of rank ten. Its standard Jacobian elliptic fibration was determined to be

$$(1) \quad \begin{aligned} y^2 = & x^3 + (u^3 + uF_4 + F_6)x^2 + (u^2G_8 + uG_{10} + G_{12})x \\ & + u^5H_8 + u^4H_{10} + u^3H_{12} + u^2H_{14} + uH_{16} + H_{18}, \end{aligned}$$

and the displayed coefficients were proven to be generators of the indicated weights for the associated ring of orthogonal modular forms. On a cover of the moduli space for Equation (1) one has $H_8 = -H_4^2$. The purpose of this article is to explain the geometric role of this weight-four character form, analogous to that of χ_{35} in higher Picard rank.

In Vinberg's scroll model, H_4 is obtained by adjoining the square root of the discriminant of two distinguished lines in a special ruling plane. Choosing its sign selects one of those lines and thereby determines an alternate, divisibility-two genus-one fibration. Taking the relative Jacobian of that alternate fibration produces the family of $U \oplus E_8(-1)$ -polarized K3 surfaces that was studied by Hashimoto and Ueda [30]. We will make every step explicit: the passage from the scroll to the standard Weierstrass equation, the neighbor transformation to the alternate quartic, and the resulting weighted-polynomial map between the Vinberg and Hashimoto–Ueda coefficient systems [66, 30, 10].

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The same character form also has an automorphic interpretation related to Borcherds' Enriques Φ -function. The occurrence of the Enriques form can be seen as follows: the invariant lattice of the K3 cover of an Enriques surface is $U(2) \oplus E_8(-2)$, and its anti-invariant lattice is

$$T_{\text{Enr}} = U \oplus U(2) \oplus E_8(-2).$$

In particular, the Borcherds form Φ_{Enr} for the Enriques modular variety is naturally a modular form on the associated type-IV domain. Dualizing and rescaling gives

$$(2) \quad T_{\text{Enr}}^{\vee}(2) \cong U^{\oplus 2} \oplus D_8(-1) = D_{2,10},$$

the period lattice of the $U \oplus D_8(-1)$ -polarized family. Since $O^+(T_{\text{Enr}}) = O^+(T_{\text{Enr}}^{\vee}(2))$, the *full* Enriques modular quotient is the same orthogonal modular variety as the full rank-ten Vinberg quotient, although their marked covers and their geometric families are different. Borcherds' weight-four Enriques form Φ_{Enr} and the character form H_4 are therefore related; their comparison requires the group, character, divisor, and cusp normalization to be fixed [5, 36, 72]. We will show that after the necessary data have been fixed, H_4 is proportional to the Borcherds–Enriques form. This connects the neighbor construction to Yoshikawa's equivariant analytic torsion and to Oberdieck's formula expressing the genus-one Enriques curve-counting series as a Fourier expansion of a power of the Enriques form [5, 36, 71, 72, 51]. In addition, it provides a physical interpretation for the form H_4 in terms of F-theory/heterotic string duality.

The lattice identity $U \oplus D_8(-1) \cong U(2) \oplus E_8(-1)$ is the lattice-theoretic source of the alternate fibration. Its relative Jacobian is $U \oplus E_8(-1)$ -polarized, and the chain rule for modular Jacobians relates the reflective character forms on the two coefficient spaces. This provides a coefficient-level connection between Vinberg's construction and the Hashimoto–Ueda ring of modular forms. Since the rank-ten family is Dolgachev self-mirror, its divisibility-one and divisibility-two fibrations are mirror to the two incident Type II boundary components, labeled by $D_8(-1)$ and $E_8(-1)$. We extract rational elliptic surfaces of type $I_4^* + 2I_1$ and $II^* + 2I_1$ from the non-K3 coefficient locus and distinguish their fiber types from the boundary lattice labels and the additional data of a polarized Type II degeneration.

We also compare the Hashimoto–Ueda coefficients with Sakai's independent Fourier–Jacobi reconstruction of the F-theory K3 equation [57]. After accounting for different normalizations, we perform a weighted change of variables to show that the eleven coefficient systems agree. This identifies Sakai's modular equation with the geometric inverse period map and relates its cusp scaling to the E-string Seiberg–Witten curve. We further identify the Witt restriction of the Hashimoto–Ueda family with the modular Inose family attached to an unordered pair of elliptic curves; its boundary divisor is exactly the Hashimoto–Ueda non-K3 locus. Finally, we restrict the construction to Kondō's five-dimensional ball quotient for eight ordered points on the projective line [37, 13, 31]. Once the finite marking data have been chosen, the restricted Vinberg coefficients become explicit homogeneous polynomial covariants of the eight-point coordinates. The factorization of H_8 is especially simple: it is a multiple of the square of a single tableau monomial.

There are two further cases that can be described using the same relative-Jacobian principle. The index-two inclusions of root lattices

$$A_7 \subset E_7, \quad A_5 \oplus A_1 \subset E_6$$

replace the role played above by $D_8 \subset E_8$. Exchanging the base and fiber variables in the A_7 and $A_5 \oplus A_1$ models and taking relative Jacobians yields explicit universal equations for K3 surfaces polarized by $U \oplus E_8(-1) \oplus A_1(-1)$ and $U \oplus E_8(-1) \oplus A_2(-1)$, respectively. The resulting coefficient weights are exactly the E_7 and E_6 weights in the free-algebra theorem of Wang–Williams [69]. Moreover, the equation

$$P_{22}^2 - 4P_{14}P_{30} = 0$$

cuts out the E_6 Heegner divisor in the E_7 modular variety and gives a geometric equation for the weight-44 Borcherds product. On the graded discriminant-kernel coefficient cone it is parametrized by $P_{14} = Q_7^2$, $P_{22} = 2Q_7Q_{15}$, and $P_{30} = Q_{15}^2$. This completes, at the level of explicit K3 equations, the exceptional E_8, E_7, E_6 cases of the ADE list of free orthogonal modular-form algebras. For reference, Appendix A collects Weierstrass equations for the K3 families associated with irreducible root systems and their Weyl-group modular forms.

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2. TWO-ELEMENTARY K3 SURFACES

Let (X, ι) be a K3 surface with a non-symplectic involution, $\iota^*\omega_X = -\omega_X$. Its invariant lattice

$$M = H^2(X, \mathbb{Z})^\iota$$

is primitive, hyperbolic, and two-elementary. Write $T_M = M^\perp \subset \Lambda_{K3}$, where

$$\Lambda_{K3} = U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}.$$

After choosing a connected component, the period domain and its discriminant arrangement are

$$\Omega_{T_M}^+ = \{[\omega] : (\omega, \omega) = 0, (\omega, \bar{\omega}) > 0\}^+, \quad \mathcal{D}_{T_M} = \bigcup_{\substack{\delta \in T_M \\ \delta^2 = -2}} \delta^\perp.$$

For the appropriate arithmetic group Γ_M , the moduli space is the open quotient

$$\mathcal{M}_M^\circ = \Gamma_M \backslash (\Omega_{T_M}^+ \setminus \mathcal{D}_{T_M}).$$

The divisor $\mathcal{D}_{T_M}$ is a Heegner divisor inside the orthogonal modular variety; its general point represents a K3 surface on which an additional (-2) -class becomes algebraic [49, 50, 72]. Write (r, ℓ, δ) for the Nikulin invariants of M . Apart from the two exceptional cases, the fixed locus of the involution is

$$(3) \quad X^\iota = C_g \sqcup R_1 \sqcup \cdots \sqcup R_k, \quad g = 11 - \frac{r + \ell}{2}, \quad k = \frac{r - \ell}{2}.$$

For $M = U(2) \oplus E_8(-2)$ it is empty, and for $M = U \oplus E_8(-2)$ it is the disjoint union of two elliptic curves.

If the fixed locus is nonempty, the quotient $q : X \rightarrow \bar{Y} = X/\iota$ is a double cover of a rational surface, branched over a divisor $B \in |-2K_{\bar{Y}}|$. Equivalently, $(\bar{Y}, \frac{1}{2}B)$ is log Calabi–Yau. In the range of the Alexeev–Nikulin correspondence, where the fixed locus contains a curve of genus at least two, the appropriate contractions in the del Pezzo–Nikulin (DPN) construction give a log del Pezzo surface S of index at most two, with branch divisor $C \in |-2K_S|$; the inverse construction is the minimal resolution of the corresponding double cover; see [1]. When the action is free, $Y = X/\iota$ is an Enriques surface. The quotient stack $[X/\iota]$ provides a uniform object: it is an Enriques surface in the free case and the square-root stack of $(\bar{Y}, B)$ in the branched case.

2.1. Yoshikawa’s analytic torsion. Yoshikawa associates to a two-elementary K3 surface of type M an equivariant analytic-torsion invariant $\tau_M(X, \iota)$. It combines the equivariant Ray–Singer torsion of X with fixed-curve and volume terms and is independent of the chosen ι -invariant Kähler metric. Let π_M denote the period map for the type- M moduli problem. His automorphicity theorem gives an integer $\nu(M) > 0$ and an automorphic form Φ_M such that

$$(4) \quad \operatorname{div}(\Phi_M) = \nu(M)\mathcal{D}_{T_M},$$

and

$$(5) \quad \tau_M(X, \iota) = \|\Phi_M(\pi_M(X, \iota))\|^{-1/(2\nu(M))}.$$

Here the group, character, and Petersson metric are those attached to the type- M moduli problem. When the fixed curve has positive genus, Φ_M is naturally an automorphic form of biweight, with values in the pullback of a Siegel Hodge bundle; it should not be confused with the pure Borchers lift used below [17, 39, 71, 72].

If s is a local equation on the period domain for a reduced component along which the divisor multiplicity of Φ_M equals $\nu(M)$, then $\Phi_M = us^{\nu(M)}$ with u a unit in a local holomorphic trivialization. If the metric remains bounded and nonvanishing, then

$$(6) \quad \log \tau_M = -\frac{1}{4} \log |s|^2 + O(1).$$

This universal leading logarithm is the precise sense in which Yoshikawa’s invariant behaves as a genus-one, one-loop determinant. The analogy with threshold corrections is useful, but (5) is the theorem used here [29, 33, 72].

The logarithmic singularity above is characteristic of a genus-one threshold singularity. Thus τ_M can be regarded as a metric-independent, automorphically completed one-loop regularized functional determinant attached to (X, ι) . We call its logarithm the genus-one B-model free energy

$$F_{1,M}^B := \log \tau_M = -\frac{1}{2\nu(M)} \log \|\Phi_M\|.$$

The divisor $\mathcal{D}_{T_M}$ is the locus where additional (-2) -classes in the anti-invariant lattice become algebraic. From the physical point of view, the divergence of τ_M near $\mathcal{D}_{T_M}$ is analogous to a one-loop threshold singularity caused by additional massless BPS states. The automorphic form Φ_M then plays a role analogous to the modular functions governing one-loop threshold corrections.

2.2. The Alexeev–Nikulin chain. The six even hyperbolic two-elementary lattices of rank ten and parity zero form the following tower:

$$\begin{aligned}
(7) \quad & U(2) \oplus E_8(-2) \longrightarrow U \oplus E_8(-2) \cong U(2) \oplus N \\
& \longrightarrow U \oplus N \cong U(2) \oplus D_4(-1)^{\oplus 2} \\
& \longrightarrow U \oplus D_4(-1)^{\oplus 2} \cong U(2) \oplus D_8(-1) \\
& \longrightarrow U \oplus D_8(-1) \cong U(2) \oplus E_8(-1) \\
& \longrightarrow U \oplus E_8(-1).
\end{aligned}$$

Here, N denotes the negative-definite Nikulin lattice, the even index-two overlattice of $A_1(-1)^{\oplus 8}$ obtained by adjoining the half-sum of the eight standard roots. Its discriminant group has length six. We denote these lattices by $M_0 = U(2) \oplus E_8(-2), \dots, M_5 = U \oplus E_8(-1)$. The discriminant length $\ell(M_i) = 10 - 2i$ decreases by 2 at each step.

The tower starts with the Enriques-cover family. Let X be the minimal resolution of a double cover of $\mathbb{P}^1 \times \mathbb{P}^1$ branched along a smooth invariant curve of bidegree $(4, 4)$ avoiding the four fixed points of the involution below, and let κ be the covering involution. The involution

$$\iota_{\text{Enr}} : (x_0 : x_1, y_0 : y_1) \longmapsto (x_0 : -x_1, y_0 : -y_1)$$

preserves the branch curve and has a lift $\tilde{\iota}_{\text{Enr}}$ of X acting trivially on the covering coordinate above its four fixed points. The composition $\iota_{\text{Enr}} := \kappa \circ \tilde{\iota}_{\text{Enr}}$ is fixed-point free. Its quotient is an Enriques surface, and its invariant lattice is isometric to $U(2) \oplus E_8(-2)$. Geometrically, a very general member admits two genus-one fibrations, one associated with each ruling, each with a distinguished bisection and no section [24, 36]. In [24], the moduli space of unpolarized Enriques surfaces is described as the quotient of the complement of the (-2) -hyperplanes in $\mathcal{D}(T_{\text{Enr}})$ by $O^+(T_{\text{Enr}})$.

Remark 2.1. The basic operation in the tower is the passage from an index-two genus-one fibration to its relative Jacobian. For $0 \leq i \leq 4$, let R_i denote the negative-definite summand in the displayed $U(2)$ -presentation of M_i . On the level of Néron–Severi lattices this replaces a primitive $U(2)$ -summand by a primitive U -summand:

$$M_i \cong U(2) \oplus R_i \quad \rightsquigarrow \quad M_{i+1} \cong U \oplus R_i.$$

For $0 \leq i \leq 3$, the resulting lattice admits a second presentation

$$U \oplus R_i \cong U(2) \oplus R_{i+1},$$

which produces the next index-two genus-one fibration in the sequence. The change between the two fibrations is described by a neighbor transformation: one replaces the original primitive isotropic fiber class by another primitive isotropic class in the same Néron–Severi lattice. In the examples considered here the two fiber classes have intersection number two, which is the source of the term 2-neighbor. Explicit models for the first transitions appear in [8]; the last two are given in Sections B and 4.

Remark 2.2. Each M_i in (7) has rank 10. For the primitive embeddings into Λ_{K3} used here, one has

$$M_i^\perp \cong U \oplus M_i.$$

Therefore every lattice in the chain is self-mirror in the sense of Dolgachev: after choosing the displayed primitive U -summand, the mirror lattice is isometric to M_i [14]. Geometrically,

the isotropic line defining the mirror cusp must still be chosen; an incident isotropic plane specifies a Type II boundary component. Zero- and one-dimensional boundary components correspond to primitive isotropic lines and planes, respectively, whereas reflective divisors in the interior record additional (-2) -classes [14, 49, 50]. Neighbor transformations alter the chosen cusp data, and relative Jacobians replace a $U(2)$ -presentation by a U -presentation.

For every M_i , Yoshikawa's theorem in [71] yields an automorphic form Φ_{M_i} with

$$\operatorname{div}(\Phi_{M_i}) = \nu(M_i)\mathcal{D}_{T_{M_i}}, \quad \text{and} \quad \tau_{M_i} = \|\Phi_{M_i}\|^{-1/(2\nu(M_i))}.$$

Yoshikawa's list in [71] is indexed by the *period lattice* L , not by the six invariant lattices M_i above. With Yoshikawa's normalization, the relevant examples are as follows. When L occurs as the orthogonal complement of an invariant K3 lattice M , unimodularity of the K3 lattice gives $\ell(L) = \ell(M)$, where $\ell(L)$ is the length of $L^\vee/L$ [72, Examples 8.8–8.13]:

- (i) For $L = U(2)^{\oplus 2} \oplus E_8(-2)$, one has $\ell(L) = 12$ and weight zero. This lattice has no primitive embedding into the K3 lattice, so the example is non-geometric for K3 surfaces; the corresponding Borcherds lift is constant [49].
- (ii) For $L = U \oplus U(2) \oplus E_8(-2)$, one has $\ell(L) = 10$ and weight four. This is the Enriques period lattice and the lift is Borcherds' Enriques Φ -function [4, 5, 6, 20, 36, 60, 71].
- (iii) For $L = U^{\oplus 2} \oplus E_8(-2)$, one has $\ell(L) = 8$ and weight twelve. The input is the theta series of the Barnes–Wall lattice divided by η^{24} , and the product is obtained by restricting Borcherds' weight-twelve form on $O(2, 26)$ [6].
- (iv) For $L = U \oplus U(2) \oplus D_4(-1)^{\oplus 2}$, one has $\ell(L) = 6$ and weight twenty-eight. This is the orthogonal model behind Kondō's moduli of eight points on $\mathbb{P}^1$. The fifteenth power of this weight-28 lift is the product of the 105 weight-four additive lifts indexed by tableaux [6, 20, 37].
- (v) For $L = U^{\oplus 2} \oplus E_8(-1)$, one has $\ell(L) = 0$ and weight 252. The Borcherds input is E_4^2/η^{24} . This is the form that also appears in the Harvey–Moore one-loop calculation and, in the coefficient model below, as the modular Jacobian of the Hashimoto–Ueda generators [6, 29, 30].
- (vi) For $L = U^{\oplus 2} \oplus D_4(-1)$, the lift has weight 72 and equals the Freitag–Hermann form, a product of 36 theta functions [19]. This lattice has rank eight; hence this last example is not a rank-twelve complement in the rank-ten tower.

Examples 8.9–8.12 correspond respectively to M_0, M_1, M_2, M_5 ; the rank-twelve complements for M_3 and M_4 do not occur in this list. The list nevertheless contains the three cases central here: the Enriques form, Kondō's eight-point form, and the weight-252 unimodular form.

These six examples concern the pure lift $\Psi_L(\cdot, F_L)$, not by themselves the complete torsion form. Here F_L is the vector-valued modular input to the Borcherds lift Ψ_L , J_M is the period map of the fixed locus to the Siegel modular variety of degree g , where g is the sum of the genera of its components, Υ_g is the corresponding Siegel modular form, and C_M is a positive normalization constant. For a non-exceptional rank-ten type (equivalently here, M_i with $i \geq 1$), the scalar factorization of Ma–Yoshikawa also contains the pulled-back Siegel factor $J_M^* \Upsilon_g$; in their normalization,

$$(8) \quad \tau_M^{-(2^g+2)(2^g-1)} = C_M \left\| \Psi_L(\cdot, (2^{g-1}+1)F_L) \right\| J_M^* \|\Upsilon_g\|.$$

For $M_1 = U \oplus E_8(-2)$, the fixed locus consists of two elliptic curves, so $g = 2$ and J_M takes values in the decomposable locus of $\mathcal{A}_2$. This distinction is important when using the examples to interpret Yoshikawa's analytic torsion [39, Theorem 9.4].

2.3. Enriques endpoint and mirror symmetry. The first lattice in the chain is

$$M_0 = M_{\text{Enr}} = U(2) \oplus E_8(-2).$$

For this lattice the involution is fixed-point free, i.e., $X^\iota = \emptyset$. The quotient $Y = X/\iota$ is an Enriques surface. Conversely, every Enriques surface arises in this way from its K3 universal cover. The invariant and anti-invariant lattices are

$$H^2(X, \mathbb{Z})^\iota \cong U(2) \oplus E_8(-2),$$

and

$$H^2(X, \mathbb{Z})^{\iota,-} \cong U \oplus U(2) \oplus E_8(-2),$$

respectively. Thus, the period domain of Enriques surfaces is the type-IV domain attached to

$$U \oplus U(2) \oplus E_8(-2).$$

In this case, Yoshikawa identifies the automorphic form $\Phi_{M_{\text{Enr}}} = \Phi_{\text{Enr}}$ with the Borchers automorphic form on the Enriques moduli space, up to a nonzero constant. Hence the B-model genus-one amplitude is governed by Borchers' Enriques Φ -function.

The A-model is the local Enriques Calabi–Yau threefold

$$X_Y^{\text{loc}} := \text{Tot}(K_Y).$$

If $\pi : X_Y^{\text{loc}} \rightarrow Y$ is the projection, then

$$K_{X_Y^{\text{loc}}} \cong \pi^*(K_Y \otimes K_Y^{-1}) \cong \mathcal{O},$$

so this is a non-compact Calabi–Yau threefold. For $g \geq 1$ and a nonzero curve class

$$\beta \in H_2(Y, \mathbb{Z})/\text{tors},$$

the local Enriques Gromov–Witten invariant is

$$N_{g,\beta} = \int_{[\overline{M}_g(Y,\beta)]^{\text{vir}}} (-1)^{g-1} \lambda_{g-1}.$$

The genus-one A-model partition function is

$$Z_{1,\text{Enr}}^A = \exp \left(\sum_{\beta \neq 0} N_{1,\beta} q^\beta \right).$$

The prediction of Klemm–Mariño [35], proved by Oberdieck in [51, Corollary 1.2], identifies this series with the Fourier expansion of a power of the Borchers automorphic form:

$$(9) \quad Z_{1,\text{Enr}}^A = \text{FE}_{\mathfrak{c}_{\text{Enr},1}} \left(\Phi_{\text{Enr}}^{-1/8} \right),$$

where $\text{FE}_{\mathfrak{c}_{\text{Enr},1}}$ denotes the Fourier expansion at the level-one cusp $\mathfrak{c}_{\text{Enr},1}$ of the Enriques moduli space. Equivalently, one may compactify the local Enriques geometry into the Enriques Calabi–Yau threefold

$$Q = (X \times E)/\langle (\iota, -1) \rangle,$$

where E is an elliptic curve. The local Enriques invariants appear as fiber-class invariants of Q . In this form the A-model can also be described by stable pairs or by Donaldson–Thomas

invariants [51]. For the broader theory of orthogonal quasimodular forms and its conjectural extensions to Enriques Gromov–Witten potentials, see [52]. Thus, the Enriques endpoint gives the following dictionary:

B-model	$\xleftrightarrow{\text{mirror symmetry}}$	A-model
Yoshikawa torsion of (X, ι)		Gromov–Witten invariants of $\text{Tot}(K_Y)$
$F_{1,\text{Enr}}^B = -\frac{1}{2\nu(M_{\text{Enr}})} \log \ \Phi_{\text{Enr}}\ .$		$Z_{1,\text{Enr}}^A = \text{FE}_{\mathbf{c}_{\text{Enr},1}}(\Phi_{\text{Enr}}^{-1/8})$

2.4. The reflective coefficient H_8 and the Enriques form. Let

$$L_{\text{Enr}} = U \oplus U(2) \oplus E_8(-2)$$

be the Enriques period lattice. The dual-rescaling isometry

$$(10) \quad L_{\text{Enr}}^\vee(2) \cong U^{\oplus 2} \oplus D_8(-1) = L_D$$

identifies the full Enriques modular quotient with the full rank-ten Vinberg quotient. Gritsenko and Nikulin identify the additive lift

$$\Delta_{4,D_8} = \text{Lift}(\vartheta(z_1) \cdots \vartheta(z_8))$$

with the Borcherds–Enriques form. Once the Weierstrass coefficient H_4 in the standard fibration (62) is normalized as this unique weight-four character generator, weight, character, and reduced divisor give

$$(11) \quad \Phi_{\text{Enr}} \doteq H_4, \quad H_8 = -H_4^2 \doteq -\Phi_{\text{Enr}}^2$$

[5, 27, 72]. The symbol $\doteq$ denotes equality up to a nonzero constant, determined by fixing a cusp coordinate and a leading Fourier coefficient.

In [51, Corollary 1.2] Oberdieck proves the genus-one Klemm–Mariño formula from [35], namely that the genus-one Enriques partition function is

$$(12) \quad Z_{1,\text{Enr}}^A := \exp\left(\sum_{\beta \neq 0} N_{1,\beta} q^\beta\right) = \prod_{\beta > 0} \left(\frac{1+q^\beta}{1-q^\beta}\right)^{a(\beta^2/2)},$$

where the coefficients $a(n)$ are defined by

$$(13) \quad \sum_{n \geq 0} a(n) q^n = \prod_{n \geq 1} \frac{(1+q^n)^8}{(1-q^n)^8} = 1 + 16q + 144q^2 + 960q^3 + 5264q^4 + \cdots.$$

In the paragraph following Corollary 1.2, he then identifies (12) with the Fourier expansion

$$(14) \quad Z_{1,\text{Enr}}^A = \text{FE}_{\mathbf{c}_{\text{Enr},1}}\left(\Phi_{\text{Enr}}^{-1/8}\right)$$

at the level-one cusp. In particular, the branch and normalization of $\Phi_{\text{Enr}}^{-1/8}$ are chosen so that its constant coefficient equals one.

We have the following:

Proposition 2.3. *After fixing the dual-rescaling identification, a sign for H_4 , and compatible local parameters at the level-one Enriques cusp and the corresponding divisibility-two Vinberg cusp, one has*

$$(15) \quad \Phi_{\text{Enr}} = c_{\text{Enr}} H_4, \quad Z_{1,\text{Enr}}^A = \text{FE}_{\mathbf{c}_{D,2}}\left((-c_{\text{Enr}}^2 H_8)^{-1/16}\right),$$

where $c_{\text{Enr}} = \frac{2}{9}$ and $\text{FE}_{\mathfrak{c}_{D,2}}$ denotes the Fourier expansion at the divisibility-two cusp of the L_D -modular variety.

Proof. Write $L_{\text{Enr}} = U_1 \oplus U_2(2) \oplus E_8(-2)$, and choose an isotropic basis e_1, f_1 of U_1 with $(e_1, f_1) = 1$. The primitive isotropic vector e_1 has divisibility one in L_{Enr} , so its line represents the level-one Enriques cusp. After dualizing and rescaling,

$$(16) \quad L_{\text{Enr}}^\vee(2) = U_1(2) \oplus U_2 \oplus E_8(-1) \cong U^{\oplus 2} \oplus D_8(-1).$$

The rational isotropic line is unchanged, but its primitive generator now has divisibility two. Thus the level-one Enriques cusp is carried to the divisibility-two Vinberg cusp:

$$(17) \quad \mathfrak{c}_{\text{Enr},1} \longmapsto \mathfrak{c}_{D,2}.$$

The associated cusp lattice is $e_1^\perp/\mathbb{Z}e_1 \cong U \oplus E_8(-1)$. The forms Φ_{Enr} and H_4 have the same weight, character, and reduced reflective divisor under the dual-rescaling identification; the one-dimensionality of this character space therefore gives $\Phi_{\text{Enr}} = c_{\text{Enr}}H_4$ for some $c_{\text{Enr}} \neq 0$ [5, 27, 10]. Finally, $H_8 = -H_4^2$ and Oberdieck's formula yield

$$\Phi_{\text{Enr}}^{-1/8} = (-c_{\text{Enr}}^2 H_8)^{-1/16},$$

which proves (15). Comparing the leading Fourier coefficients in the two chosen cusp coordinates gives $c_{\text{Enr}} = \frac{2}{9}$. $\square$

Remark 2.4. In the standard fibration (1), the divisor $H_8 = 0$ generically enhances the I_4^* -fiber to I_9^* ; hence the lattice polarization extends from $U \oplus D_8(-1)$ to a lattice containing $U \oplus D_9(-1)$. On the character cover, $H_8 = -H_4^2$, so H_4 is a reduced local equation of this reflective divisor [10]. Under heterotic/F-theory duality, the complex-structure moduli of an elliptically fibered K3 surface are identified with heterotic Narain moduli. Gauge enhancement occurs when the Narain lattice acquires additional norm-two vectors with vanishing right-moving momentum. Correspondingly, the one-loop heterotic threshold has logarithmic singularities along the rational quadratic divisors

$$p_R = 0, \quad p_L^2 = 2.$$

The divisor of Φ_{Enr} identifies this enhancement locus in the orthogonal period domain [29, 33].

2.5. Sakai's Fourier–Jacobi reconstruction and the E-string limit. Sakai has recently given an independent construction of the modular coefficients of the Hashimoto–Ueda equation from the Fourier–Jacobi expansions of the orthogonal Eisenstein series [57]. His motivation comes from eight-dimensional heterotic/F-theory duality: the period variables (τ, ρ, z) describe the complex and complexified Kähler moduli of the heterotic two-torus together with eight Wilson-line parameters in one E_8 factor, while the dual F-theory background is an elliptically fibered K3 surface with one fiber of type II^* [64, 47, 48, 11].

We first compare his normalization with ours. Let $\mathcal{E}_k$ denote the normalized Eisenstein series used in this article, whose zeroth Fourier–Jacobi coefficient is the elliptic Eisenstein series $E_k(\tau)$ normalized to have constant Fourier coefficient 1. If $\mathcal{E}_k^{\text{S}}$ denotes Sakai's Eisenstein series, then his Equation (4.7) reads

$$(18) \quad \mathcal{E}_k^{\text{S}}(\tau, \rho, z) = -\frac{B_k}{2k} E_k(\tau) + \sum_{m \geq 1} (E_{k-4}(\tau) \Theta(\tau, z)) |T(m) p^m, \quad p = e^{2\pi i \rho}.$$

Here B_k is the k th Bernoulli number, Θ is the E_8 theta function, $T(m)$ is the Jacobi index-raising Hecke operator, and $E_0 = 1$. Consequently,

$$(19) \quad \mathcal{E}_k^S = -\frac{B_k}{2k} \mathcal{E}_k, \quad \mathcal{E}_k = -\frac{2k}{B_k} \mathcal{E}_k^S.$$

The numerical conversion factors $-\frac{2k}{B_k}$ from Sakai's normalization to ours begin with

$$240, \quad -264, \quad \frac{65520}{691} \quad (k = 4, 10, 12).$$

Sakai writes the F-theory background as a Jacobian elliptic K3 surface with Weierstrass equation

$$(20) \quad \begin{aligned} y_S^2 = & 4x_S^3 - (\varphi_4 u^4 + \varphi_{10} u^3 + \varphi_{16} u^2 + \varphi_{22} u + \varphi_{28}) x_S \\ & + 4u^7 - (\varphi_{12} u^5 + \varphi_{18} u^4 + \varphi_{24} u^3 + \varphi_{30} u^2 + \varphi_{36} u + \varphi_{42}). \end{aligned}$$

Each φ_k is taken to be the most general weighted polynomial of weight k in the eleven Eisenstein generators; altogether this ansatz contains 76 rational constants. After the translation

$$u \mapsto u - \frac{\varphi_{10}}{4\varphi_4},$$

we write $\tilde{\varphi}_k$ for the resulting coefficients. Sakai imposes the vanishing conditions

$$(21) \quad \begin{aligned} \tilde{\varphi}_4 &= a_0 + O(p), \quad \tilde{\varphi}_{4+6i} = O(p^i) \quad (i = 2, 3, 4), \\ \tilde{\varphi}_6 &= b_0 + O(p), \quad \tilde{\varphi}_{6+6j} = O(p^j) \quad (j = 1, \dots, 6), \end{aligned}$$

where $a_0 = E_4/12$ and $b_0 = E_6/216$. The coefficient of p^m in an orthogonal modular form is a $W(E_8)$ -invariant Jacobi form of index m . Expressing these coefficients in the algebraically independent meromorphic Jacobi generators a_i, b_j turns (21) into an overdetermined *linear* system for the 76 constants. Algebraic independence allows coefficients of distinct monomials in the a_i, b_j to be compared, and the system has a unique solution. This is the uniqueness step in Sakai's argument. Note that no uniqueness of K3 periods is assumed here. See [56, 57, 58, 68] for background on E_8 Jacobi forms.

The solution satisfies the sharper leading-term identities

$$(22) \quad \tilde{\varphi}_{4+6n} = a_n \Delta^n p^n + O(p^{n+1}), \quad \tilde{\varphi}_{6+6n} = b_n \Delta^n p^n + O(p^{n+1}).$$

Here $n \in \{0, 2, 3, 4\}$ in the first identity and $n \in \{0, 1, \dots, 6\}$ in the second. After

$$(23) \quad u \mapsto (\Delta p)u, \quad x \mapsto (\Delta p)^2 x, \quad y \mapsto (\Delta p)^3 y,$$

and division by the common factor, the limit $p \rightarrow 0$ is the rational elliptic surface

$$(24) \quad \begin{aligned} y^2 = & 4x^3 - (a_0 u^4 + a_2 u^2 + a_3 u + a_4) x \\ & - (b_0 u^6 + b_1 u^5 + b_2 u^4 + b_3 u^3 + b_4 u^2 + b_5 u + b_6), \end{aligned}$$

the Seiberg–Witten curve of the E-string theory [15, 16, 43, 55]; for the six-dimensional origin of the E-string, see [22, 23, 61]. In F-theory language, the K3 discriminant accounts for 24 seven-branes. The fixed II^* fiber uses ten units of discriminant multiplicity; among the other fourteen branes, the scaling keeps twelve in the finite region and moves two to infinity, thereby isolating a rational elliptic surface. A D3-brane probing this limiting background realizes the E-string theory compactified on T^2 . If one instead sets $p = 0$ without the rescaling (23),

one obtains a non-minimal Weierstrass model in the excluded, non-K3 locus. This is the same boundary phenomenon that appears in the Witt–Inose restriction in Theorem 6.2.

The remaining issue in [57] is as follows: the degeneration conditions determine a unique tuple of orthogonal modular forms, but by themselves do not identify that tuple with the geometric inverse of the K3 period map. However, the latter identification follows by comparison with the Hashimoto–Ueda normal form:

Proposition 2.5. *Let φ_k^S be Sakai’s eleven coefficient polynomials, written in the Eisenstein series $\mathcal{E}_k^S$, and let C_k be the Hashimoto–Ueda coefficients in the normalization of (155). Then, after the Eisenstein normalization (19), one has*

$$(25) \quad C_k = -\frac{1}{4}(-12)^{k/2}\varphi_k^S, \quad k \in \{4, 10, 12, 16, 18, 22, 24, 28, 30, 36, 42\}.$$

Consequently, Sakai’s modular equation is, up to the weighted coordinate change, the Hashimoto–Ueda universal family. In particular, it gives the inverse period map on the common dense open moduli locus, as conjectured in [57].

Proof. Choose $\lambda \in \mathbb{C}^\times$ with $\lambda^2 = -12$ and make the weighted change of variables

$$(26) \quad t = \lambda^6 u, \quad x = \lambda^{14} x_S, \quad y = \frac{\lambda^{21}}{2} y_S.$$

Dividing (20) first by 4 and then applying (26) gives a monic equation with leading term t^7 and sends every coefficient of weight k to $-\frac{1}{4}\lambda^k\varphi_k^S$. Since $\lambda^k = (-12)^{k/2}$, this proves the formal coordinate relation in (25).

It remains to identify the modular forms. Use the compatible period coordinates of Lemma 4.4, with the common cusp coming from the displayed $U^{\oplus 2}$ in $T_D \subset T_E$, and put $p = e^{2\pi i \rho}$. The geometric relative-Jacobian coefficients are holomorphic modular forms of the indicated weights for $O^+(T_E)$ by the period identification and the Hashimoto–Ueda theorem. The fact that the graded ring of modular forms is generated by Eisenstein series [12, Theorem 4.3] places the coefficients in Sakai’s weighted-polynomial ansatz. In the normalization of the Vinberg coefficient forms, [10, Section 5 and Corollary 5.1] gives

$$\begin{aligned} F_4 &= -3E_4 + O(p), & F_6 &= 2E_6 + O(p), \\ G_8, G_{10}, G_{12}, H_4 &= O(p), & H_{10}, \dots, H_{18} &= O(p^2). \end{aligned}$$

Here the assertion for H_4 follows from $H_8 = -H_4^2 = O(p^2)$. Equation (71) now shows that $A_i(pS) = O(p^2)$ for every i . Since the binary-quartic invariants are homogeneous of degrees two and three in the A_i , respectively, writing

$$f(s) := -\frac{1}{3}\mathcal{I}(s) = \sum_{j=0}^4 f_j(p)s^j, \quad g(s) := -\frac{1}{27}\mathcal{J}(s) = s^7 + \sum_{j=0}^6 g_j(p)s^j$$

gives

$$\begin{aligned} f_j(p) &= O(p^{4-j}) \quad (0 \leq j \leq 4), \quad f_4(0) = -3E_4, \\ g_j(p) &= O(p^{6-j}) \quad (0 \leq j \leq 6), \quad g_6(0) = 2E_6. \end{aligned}$$

Indeed, at $p = 0$ the equation is $y^2 = x^3 + F_4 s^4 x + s^6(s + F_6)$. On the dense open set $E_4 \neq 0$, the translation $s = S - f_3/(4f_4)$ has shift $O(p)$, eliminates the cubic term in f , and preserves all of the above order bounds. Since $t = s + F_6/7$, this is exactly $t = S - C_{10}/(4C_4)$, the

translation used by Sakai after the weighted coordinate change. In Sakai's scale, the leading quartic and sextic coefficients are consequently

$$-4\lambda^{-4}(-3E_4) = \frac{E_4}{12} = a_0, \quad -4\lambda^{-6}(2E_6) = \frac{E_6}{216} = b_0.$$

The geometric coefficients therefore satisfy all of (21). The uniqueness of Sakai's solution to these conditions [57, Section 5] proves (25); equality extends over the whole period domain by holomorphy.

Substituting (19) into Sakai's Appendix A gives the explicit Eisenstein expressions. For example, the first four identities become

$$\begin{aligned} C_4 &= -3\mathcal{E}_4, & C_{10} &= \frac{24}{7}\mathcal{E}_{10}, \\ C_{12} &= \frac{378}{125}\mathcal{E}_4^3 - \frac{4146}{875}\mathcal{E}_{12}, & C_{16} &= -\frac{10557}{3125}\mathcal{E}_4^4 + \frac{3756276}{284375}\mathcal{E}_4\mathcal{E}_{12} - \frac{22494123}{1990625}\mathcal{E}_{16}. \end{aligned}$$

The complete list of eleven identities is given in Appendix G. Hashimoto–Ueda's weighted coefficient space, with the non-K3 locus removed, is the coarse moduli space of the corresponding $U \oplus E_8(-1)$ -polarized K3 surfaces. Hence equality of the weighted coefficient maps identifies Sakai's map with the geometric inverse period map, up to the coordinate scaling and the finite marking ambiguity inherent in the coarse quotient. $\square$

3. SELF-MIRROR FIBRATIONS AND TYPE II BOUNDARY COMPONENTS

The two genus-one fibrations used in the transition from Vinberg's $U \oplus D_8(-1)$ -polarized family to the Hashimoto–Ueda $U \oplus E_8(-1)$ -polarized family also have an interpretation in Dolgachev's lattice-theoretic mirror symmetry for K3 surfaces.

3.1. The mirror lattice at a cusp. For a primitive hyperbolic lattice embedding into the K3 lattice $M \subset \Lambda_{K3}$, Dolgachev's mirror symmetry procedure starts with a choice of a primitive isotropic vector of divisibility one in $M^\perp$. This results in a splitting of a hyperbolic plane $M^\perp \cong U \oplus \check{M}$, with the remaining summand $\check{M}$ viewed as the *mirror lattice*. The construction depends on the choice of isotropic vector. This vector also determines a zero-dimensional *boundary component* (cusp) in the Baily–Borel compactification of the arithmetic period quotient associated to $M^\perp$.

For the rank-ten family considered here, the polarizing lattice M has the special property that, for an appropriate choice of Baily–Borel cusp, the Dolgachev mirror lattice $\check{M}$ is isomorphic to the original M . Let us then set

$$(27) \quad M_D = U \oplus D_8(-1), \quad T_D := M_D^\perp \subset \Lambda_{K3}, \quad T_D \cong U_0 \oplus M_D.$$

Thus M_D is its own Dolgachev mirror after the copy U_0 has been chosen [14]. For the boundary-orbit count below, we use the full orthogonal quotient $O^+(T_D) \backslash \mathcal{D}(T_D)$. Fix a primitive isotropic vector $e \in U_0$. Then $\mathbb{Q}e$ determines a zero-dimensional cusp in its Baily–Borel compactification. A one-dimensional Baily–Borel boundary component incident to this cusp is represented by a primitive isotropic plane $I \subset T_D$ containing e . Modulo $\mathbb{Z}e$, such a plane determines a primitive isotropic line $\mathbb{Z}f \subset e^\perp/\mathbb{Z}e \cong \check{M} \cong M_D$. After choosing a nef representative of f , this gives a genus-one fibration on the mirror K3 surface. In our self-mirror family, the lattice-isometry types are described by the following lemma:

Lemma 3.1. *The primitive isotropic vectors in $L = U \oplus D_8(-1)$ form exactly two orbits under $O(L)$, distinguished by divisibility one and two. Their orthogonal quotients are $D_8(-1)$ and $E_8(-1)$, respectively.*

Proof. Choose a basis $(p, q, d_1, \dots, d_8)$, where $p^2 = q^2 = 0$, $(p, q) = 1$, and the d_i have the negative D_8 Cartan matrix: the chain is $d_1, \dots, d_6$, with d_7, d_8 attached to d_6 . Vinberg's algorithm, as implemented in VinAl [3], gives the ten facet normals

$$\begin{aligned} r_i &= -d_i \quad (1 \leq i \leq 7), \quad r_8 = d_7 - d_8, \\ r_9 &= p - q, \quad r_{10} = q + d_1 + 2d_2 + 2d_3 + 2d_4 + 2d_5 + 2d_6 + d_7 + d_8. \end{aligned}$$

The output can be checked directly. All ten reflections preserve L ; $r_8^2 = -4$ and the other normals have square -2 . Their nonright dihedral angles are $\pi/3$ and $\pi/4$. The dual vectors defined by $(z_i, r_j) = -\delta_{ij}$ lie in the closure of the same positive cone. Their squared norms, in order, are

$$(z_1^2, \dots, z_{10}^2) = (1, 6, 5, 4, 3, 2, 1, 0, 0, 2).$$

Thus the chamber is a finite-volume Coxeter simplex with ten facets, eight finite vertices, and two ideal vertices. The ideal rays are represented, in the chosen basis, by

$$\begin{aligned} v_1 &= (0, -1, 0, 0, 0, 0, 0, 0, 0, 0), \\ v_2 &= (-1, -1, -\tfrac{1}{2}, -1, -\tfrac{3}{2}, -2, -\tfrac{5}{2}, -3, -\tfrac{3}{2}, -2). \end{aligned}$$

The reflection group therefore has two cusp orbits in the chosen positive cone, so $O(L)$ has at most two primitive isotropic orbits. The primitive integral generators are $f_1 = v_1$ and $f_2 = 2v_2$; direct pairing with the basis gives $\text{div}(f_1) = 1$, $\text{div}(f_2) = 2$, and $(f_1, f_2) = 2$. Divisibility prevents their identification, while $-\text{id}$ identifies opposite rays. Hence there are exactly two orbits. The first quotient is visibly $D_8(-1)$. For the second, reflection in r_8 sends f_2 to $g = f_2 - r_8$. Both have divisibility two, and $(f_2, g) = 2$, so their span splits off integrally as $U(2)$. Its complement is even, negative definite of rank eight, and unimodular, and is therefore $E_8(-1)$. This also proves the alternative presentation $L \cong U(2) \oplus E_8(-1)$. $\square$

Let X be very general, so that $\text{NS}(X) = M_D$. For the orbit of divisibility one, the fiber and zero-section classes span a copy of U in M_D . Their orthogonal complement is $D_8(-1)$, which is generated by the components of the distinguished I_4^* fiber that do not meet the zero section. The Shioda–Tate formula gives Mordell–Weil rank zero, and comparing discriminants shows that the torsion subgroup is trivial. This is the *standard* Weierstrass model (1); generically it has fibers $I_4^* + 14I_1$ and trivial Mordell–Weil group.

For the orbit of divisibility two, the fiber class f has divisibility two in $\text{NS}(X)$. The old fiber has intersection two with f and gives a bisection, while a section would contradict $\text{div}(f) = 2$. The multisection index is therefore two, and the generic fiber is a torsor of order two under its Jacobian [42]. In Proposition 4.3 we will construct the torsor explicitly, as the *alternate* genus-one fibration given by the binary-quartic family

$$(28) \quad w^2 = A_0(s)u^4 + A_1(s)u^3 + A_2(s)u^2 + A_3(s)u + A_4(s).$$

It has a bisection and generically no section. The relative Jacobian (75) has a section and contains $U \oplus E_8(-1)$ in its Néron–Severi lattice; generically its distinguished singular fiber is II^* . Thus its relative Jacobian is the Hashimoto–Ueda fibration.

We have the following:

Proposition 3.2. *Fix the cusp $\mathbb{Q}e$ and the full orthogonal quotient as above. The two $O(M_D)$ -orbits of primitive isotropic vectors correspond to two distinct one-dimensional Baily–Borel boundary components incident to this cusp. On a very general M_D -polarized K3 surface X , their nef representatives give the following types of genus-one fibrations:*

	$\text{div}(f)$	presentation of M_D	$f^\perp/\mathbb{Z}f$	fibration on X
(29)	1	$U \oplus D_8(-1)$	$D_8(-1)$	Jacobian elliptic
	2	$U(2) \oplus E_8(-1)$	$E_8(-1)$	index-two genus one.

The divisibility-one fibration is the standard elliptic fibration (1); the divisibility-two fibration is the alternate genus-one fibration (28).

Proof. Rational boundary components of a type-IV quotient are indexed by arithmetic orbits of isotropic lines and planes: the former give zero-dimensional cusps and the latter one-dimensional Type II components [59]. An isotropic plane incident to $\mathbb{Q}e$ projects through $e^\perp/\mathbb{Z}e \cong M_D$ to a primitive isotropic line $\mathbb{Z}f \subset M_D$. The cusp stabilizer in the full orthogonal group induces the full isometry action on these lines, using the splitting $T_D = U_0 \oplus M_D$. Lemma 3.1 therefore gives exactly two incident boundary components, which are distinguished by their nonisometric quotients $I^\perp/I$. After changing sign if necessary and acting by the (-2) -reflection group, f may be taken nef. A primitive nef class of square zero on a K3 surface defines a genus-one fibration, and the divisibility of f is its index. Thus divisibility one gives a section, whereas divisibility two gives a bisection and no section on the very general member. Finally,

$$U \oplus D_8(-1) \cong U(2) \oplus E_8(-1),$$

and the fiber vectors in the two displayed decompositions represent the two orbits of Lemma 3.1. Taking the orthogonal quotient by the corresponding fiber class gives $D_8(-1)$ and $E_8(-1)$, respectively. The asserted Kodaira configurations and Mordell–Weil groups follow from the explicit models and the Shioda–Tate formula; see Sections 4 and C. $\square$

3.2. Type II stable limits. We next relate the Baily–Borel labels to geometric Type II limits. A Kulikov model of Type II has smooth elliptic double curves. In an elliptic stable degeneration one can encounter a central fiber of the form

$$(30) \quad X_0 = R_1 \cup_E R_2,$$

where R_1 and R_2 are rational elliptic surfaces meeting along a common smooth fiber E ; see [38, 54, 21, 11]. The Baily–Borel compactification retains the modular parameter of the elliptic double curve E but contracts the additional extension data present in a toroidal compactification.

We will compute the following specializations of the standard and alternate pencils directly from the two irreducible components of the non-K3 locus computed in [10, Section 2.4 and Proposition 2.15]. To distinguish these curves from the modular coefficients C_k , we write

$$\mathcal{C}_D := C_1, \quad \mathcal{C}_E := C_2$$

for the components denoted by C_1 and C_2 there. The subscripts will be explained below. The residual surfaces after specialization will be extremal rational elliptic surfaces with fiber root lattices E_8 and D_8 , respectively [44, 45, 53]. For any rational elliptic surface R with section and fiber class F_R , one has $F_R^\perp/\mathbb{Z}F_R \cong E_8(-1)$ in its Néron–Severi lattice [44], and the labels do *not* refer to the fiber root lattices of the rational elliptic surfaces. Varying

a chosen smooth fiber on either surface supplies an elliptic modulus; identifying this with boundary data requires the chosen marking and degeneration.

The first component of the non-K3 locus is

$$(31) \quad \mathcal{C}_D = V(G_8, G_{10}, G_{12}, H_8, H_{10}, H_{12}, H_{14}, H_{16}, H_{18}) \cong \mathbb{P}(4, 6),$$

with coordinates $[F_4 : F_6]$; this is Equation (2.16) of [10]. The second component $\mathcal{C}_E$ is defined by the seventeen weighted homogeneous generators displayed in [10, Lemma 2.13 and Appendix B, Equations (B.1)–(B.4)]. On the character cover on which H_4 is defined, a parametrization by $[\alpha : F_4] \in \mathbb{P}(2, 4)$ is

$$(32) \quad \begin{aligned} H_4 &= \alpha^2 + \frac{1}{3}F_4, & F_6 &= 10\alpha^3 + 4\alpha F_4, \\ G_8 &= \frac{1}{3}F_4^2 - 15\alpha^4 - 4\alpha^2 F_4, & G_{10} &= -12\alpha^5 + 4\alpha^3 F_4 + \frac{8}{3}\alpha F_4^2, \\ G_{12} &= 30\alpha^6 + 26\alpha^4 F_4 + \frac{16}{3}\alpha^2 F_4^2, & H_8 &= -\alpha^4 - \frac{2}{3}\alpha^2 F_4 - \frac{1}{9}F_4^2, \\ H_{10} &= -8\alpha^5 - \frac{16}{3}\alpha^3 F_4 - \frac{8}{9}\alpha F_4^2. \end{aligned}$$

The remaining coefficients $H_{12}, \dots, H_{18}$ are given in [10, Equation (2.19)]; in particular, we have $H_8 = -H_4^2$. The boundary lattice labels are then determined as follows: at the standard one-dimensional cusp of $T_D = 2U \oplus D_8(-1)$, the Fourier–Jacobi expansions of F_4, F_6 , of G_8, G_{10}, G_{12} , and of $H_8, \dots, H_{18}$ begin in orders zero, one, and two, respectively [10, Corollary 5.1]. Thus, the boundary value lies on $\mathcal{C}_D$, with coordinates $[F_4 : F_6]$, and $\mathcal{C}_D$ has orthogonal quotient $D_8(-1)$. The other boundary curve $\mathcal{C}_E$ has quotient $E_8(-1)$. That is, the labels refer to the quotients $I_{T_D}^\perp / I$ of the period lattice.

Starting with $\mathcal{C}_E$, we convert the standard fibration (1) to short Weierstrass form and substitute (32). One obtains

$$(33) \quad \begin{aligned} Y^2 &= X^3 - \frac{1}{3}(u + \alpha)^4(u^2 - 4\alpha u + 10\alpha^2 + 2F_4)X \\ &\quad - \frac{2}{27}(2\alpha - u)(u + \alpha)^6(u^2 - 4\alpha u + 13\alpha^2 + 3F_4). \end{aligned}$$

For general $[\alpha : F_4]$, the Weierstrass equation of the standard fibration acquires a non-minimal $(4, 6, 12)$ point at $u = -\alpha$ on the base. We set $d = u + \alpha$, and the substitution

$$(34) \quad X = d^2 X_1, \quad Y = d^3 Y_1$$

removes the common $(4, 6, 12)$ factor and produces the rational elliptic surface. We make the constant rescaling $Z = 3X_1$, $\eta = 3\sqrt{3}Y_1$, followed by the translation $Z = \xi + u - 2\alpha$. The resulting rational elliptic surface is

$$(35) \quad R : \quad \eta^2 = \xi^3 + 3(u - 2\alpha)\xi^2 - 6(3\alpha^2 + F_4)\xi.$$

After setting

$$v = 3u, \quad a = -6\alpha, \quad b = -6(3\alpha^2 + F_4),$$

this becomes the rational elliptic surface

$$(36) \quad R : \quad y^2 = x^3 + (v + a)x^2 + bx, \quad \Delta = 16b^2((v + a)^2 - 4b),$$

with $b \neq 0$. For general $[\alpha : F_4]$, the singular fibers are $I_4^* + 2I_1$ and $\text{MW}(R) \cong \mathbb{Z}/2\mathbb{Z}$. The section $(x, y) = (0, 0)$ generates the order-two Mordell–Weil group. Thus the standard pencil keeps the I_4^* fiber in the residual equation (36); see [10, Remark 2.14].

Next we consider $\mathcal{C}_D$. Substitution of (31) into the standard Weierstrass equation gives

$$(37) \quad y^2 = x^2(x + u^3 + F_4u + F_6),$$

which is singular along the curve $x = y = 0$; see [10, Lemma 2.11 and Remark 2.12]. The original projection to the u -line has singular generic fiber. Instead, regard x as the base coordinate and put $Y = y/x$ to remove the square factor. This gives

$$(38) \quad Y^2 = u^3 + F_4u + (x + F_6)$$

as the residual rational elliptic surface. In the monic Weierstrass model $V^2 = U^3 + F_4x^4U + x^6(x + F_6)$, with $U = x^2u$ and $V = x^2y$, this removes a non-minimal $(4, 6, 12)$ point at $x = 0$ for general parameters. On $\mathcal{C}_D$ one has $H_4 = 0$, so the alternate base coordinate $s = x - H_4u$ specializes to $s = x$. Thus the change of ruling above is precisely the specialization of the alternate pencil. With $s = t - F_6/7$, its equation factors as

$$(39) \quad w^2 = \frac{1}{343}(7t - F_6)^2(7u^3 + 7F_4u + 7t + 6F_6).$$

Removing the square factor by $y = 7w/(7t - F_6)$ gives the same surface, with $x = t - F_6/7$. As an elliptic surface over the v -line, we set $A = F_4$, $v = x$, $B = F_6$ and rename the fiber coordinates (u, Y) as (x, y) to obtain

$$(40) \quad R: \quad y^2 = x^3 + Ax + (v + B), \quad \Delta = -16(4A^3 + 27(v + B)^2).$$

The general member has fibers $\text{II}^* + 2I_1$ and trivial Mordell–Weil group.

With f_{src} denoting the fiber class of the pencil being specialized, we have obtained the following residual rational elliptic surfaces:

	$\text{div}(f_{\text{src}})$	coefficient curve	pencil specialized	$\text{Fib}(R)$	$\text{MW}(R)$
(41)	1	$\mathcal{C}_E$	standard	$I_4^* + 2I_1$	$\mathbb{Z}/2\mathbb{Z}$
	2	$\mathcal{C}_D$	alternate	$\text{II}^* + 2I_1$	0

Note that Proposition 3.2 instead associates a boundary plane $I \subset T_D$ with a fiber class on the mirror K3 surface. Self-mirror lattice isometry does not identify that mirror fiber class with f_{src} in the specialization table.

This construction is compatible with its F-theory interpretation. An elliptically fibered K3 surface describes a type-IIB background on $\mathbb{P}^1$ with 24 units of seven-brane charge. The stable degeneration (30) separates the background into two twelve-brane blocks, each carried by a rational elliptic surface, and the common fiber E is the elliptic curve of the dual heterotic torus [11, 47, 48, 64]. The boundary labels $D_8(-1)$ and $E_8(-1)$ encode the two mirror fibration types; the direct specializations of the original pencils are those in (41). These extractions do not specify both twelve-brane blocks of a polarized stable degeneration. On the mirror K3 side, the two lattice labels are realized by the standard Jacobian fibration and the alternate index-two fibration, whose relative Jacobian is of Hashimoto–Ueda type. This is the boundary counterpart of the neighbor construction carried out in the next section, and it is compatible with the CHL interpretation of the $U(2)$ -polarized genus-one model [8].

4. FROM VINBERG'S CONSTRUCTION TO THE WEIERSTRASS MODEL

We will now describe the explicit relation between Vinberg's projective model of the rank-ten family and the standard and alternate genus-one fibrations supported on the family of K3 surfaces polarized by the lattice $U \oplus D_8(-1)$. The geometric construction can be summarized as follows:

$$(42) \quad Y_V^{\min} \dashrightarrow X_{F,G,H} \dashrightarrow \mathcal{C}/\mathbb{P}_{(s)}^1 \xrightarrow{\text{Jac}} X_{\text{HU}}.$$

The first arrow identifies Vinberg's scroll model in [66] with (62); the second changes the genus-one pencil on the same K3 surface; the last takes the relative Jacobian. Hashimoto and Ueda showed that the ring of scalar modular forms corresponding to the latter family is polynomial in eleven generators and that adjoining the weight-252 character form gives the full ring [30].

4.1. Vinberg multipolarizations and arithmetic groups. Let $h_0 \in \Lambda_{K3}$ be primitive with $h_0^2 = d > 0$, and let $S_0 \subset \Lambda_{K3}$ be a primitive hyperbolic lattice containing h_0 . Put $T_0 = S_0^\perp$. In [65, Section 4.1], a multipolarization of type (h_0, S_0) on a K3 surface X consists of a class h in the chosen nef chamber and an *unlabelled* sublattice $S \subset \text{NS}(X)$ containing h , such that some marking

$$\varphi : H^2(X, \mathbb{Z}) \xrightarrow{\sim} \Lambda_{K3}$$

sends (h, S) to (h_0, S_0) . The marking itself is not part of the object. It is defined only up to an isometry of Λ_{K3} preserving h_0 and S_0 . Vinberg then defines

$$\text{O}^+(T_0, h_0) = \text{res}_{T_0} \{g \in \text{O}^+(\Lambda_{K3}) : g(h_0) = h_0, \quad g(S_0) = S_0\}.$$

The period space is

$$\text{O}^+(T_0, h_0) \backslash \mathcal{D}(T_0),$$

up to the hyperplane arrangements needed to impose ampleness or exclude extra algebraic classes. Vinberg states that $\text{O}^+(T_0, h_0)$ has finite index in $\text{O}^+(T_0)$.

For an even lattice K , write

$$A_K = K^\vee/K, \quad q_K : A_K \longrightarrow \mathbb{Q}/2\mathbb{Z}$$

for its discriminant form, and set

$$\tilde{\text{O}}(K) = \ker(\text{O}(K) \longrightarrow \text{O}(q_K)).$$

This is the stable orthogonal group or discriminant kernel. Because Λ_{K3} is unimodular, a primitive decomposition $S_0 \perp T_0 \subset \Lambda_{K3}$ determines an anti-isometry

$$\lambda : (A_{S_0}, q_{S_0}) \xrightarrow{\sim} (A_{T_0}, -q_{T_0}).$$

Vinberg's extension criterion [65, Proposition 4] states that $(g_S, g_T) \in \text{O}(S_0) \times \text{O}(T_0)$ extends to the K3 lattice precisely when

$$\bar{g}_T \circ \lambda = \lambda \circ \bar{g}_S.$$

Consequently, we introduce

$$(43) \quad \Gamma_{S_0, h_0}^+ = \left\{ g_T \in \text{O}^+(T_0) : \begin{array}{l} \exists g_S \in \text{O}(S_0) : g_S(h_0) = h_0, \\ \bar{g}_T = \lambda \bar{g}_S \lambda^{-1} \end{array} \right\},$$

which is Vinberg's $\text{O}^+(T_0, h_0)$, and we have

$$(44) \quad \tilde{\text{O}}^+(T_0) \subseteq \Gamma_{S_0, h_0}^+ \subseteq \text{O}^+(T_0).$$

The three groups have distinct moduli meanings:

- (1) if the primitive embedding $S_0 \hookrightarrow \text{NS}(X)$ is fixed pointwise, the period group is $\tilde{\text{O}}^+(T_0)$;
- (2) if only the image lattice and h_0 are remembered, the period group is Γ_{S_0, h_0}^+ ;
- (3) if one also forgets h_0 , all of $\text{O}(S_0)$ is allowed. When the reduction maps onto the discriminant-form groups are surjective, the resulting group on T_0 is the full $\text{O}^+(T_0)$.

If π_T and π_{S, h_0} denote the corresponding homomorphisms to the discriminant-form groups, then

$$\Gamma_{S_0, h_0}^+ / \tilde{\text{O}}^+(T_0) \cong \text{im}(\pi_T) \cap \lambda \text{im}(\pi_{S, h_0}) \lambda^{-1},$$

with the component-preserving condition understood. In particular, if π_T is surjective, a multipolarized quotient is the full quotient exactly when the stabilizer of h_0 supplies every discriminant action; compare [24].

Vinberg's actual choices in [65] are

$$S_n = D_{1, 19-n}, \quad T_n = D_{2, n}, \quad 3 \leq n \leq 7,$$

together with a particular vector $h_0 \in S_n$ of square 4. From [65, Prop. 7] we have

$$\text{O}^+(T_n, h_0) = \begin{cases} \Gamma_7^0, & n = 7, \\ \Gamma_6^{\text{ext}} = \text{O}^+(T_6), & n = 6, \\ \Gamma_n = \text{O}^+(T_n), & n = 5, 4, 3. \end{cases}$$

In the case $n = 7$, the multipolarized quotient is not the full quotient. Vinberg's canonical equations are quartics in $\mathbb{P}^3$ for these S_n . The Enriques lattice, whose discriminant has absolute value 2^{10} , does not occur. In [66], Vinberg considers the rank-ten case with

$$D_{1, 9} \cong U \oplus D_8(-1)$$

and a distinguished polarization of square eight. The resulting canonical scroll model is discussed in Section 4.2.

4.2. Vinberg's canonical scroll model. Let

$$\mathcal{S}_{\text{sc}} = \mathcal{S}(0, 1, 2) \subset \mathbb{P}^5$$

be the singular rational normal scroll with homogeneous coordinates

$$(x_0, x_1, x_2, y_1, y_2, z).$$

Its standard resolution is the projective bundle

$$\tilde{\mathcal{S}}_{\text{sc}} := \mathbb{P}_{\mathbb{P}^1}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(2)).$$

The tautological morphism $\tilde{\mathcal{S}}_{\text{sc}} \rightarrow \mathcal{S}_{\text{sc}}$ contracts the section associated with the trivial summand to the vertex $[1 : 0 : 0 : 0 : 0 : 0]$. Its fibers map to a one-parameter family of ruling planes $L(u) \cong \mathbb{P}^2$, all meeting at that vertex. The image is a threefold of degree three. In the homogeneous coordinates $(x_0, x_1, x_2, y_1, y_2, z)$, its homogeneous ideal is generated by

$$(45) \quad x_2 y_1 = x_1 z, \quad x_1 y_2 = x_2 z, \quad y_1 y_2 = z^2.$$

On an affine chart of the base, the ruling plane over $u \in \mathbb{A}^1$ is parametrized by

$$x_2 = u x_1, \quad z = u y_1, \quad y_2 = u^2 y_1,$$

with $(x_0 : x_1 : y_1)$ serving as homogeneous coordinates on $L(u)$.

Vinberg's canonical cubic is

$$\begin{aligned}
(46) \quad \mathcal{F} = & x_0^2 y_1 + x_1^3 + x_2^2 z + (ax_1 + by_1)y_1^2 \\
& + (a_1 x_1 y_1 + a_2 x_2 y_2 + b_1 y_1^2 + b_2 y_2^2)z \\
& + (f_1 x_1 + f_2 x_2 + g_1 y_1 + g_2 y_2 + hz)z^2.
\end{aligned}$$

Restricting Vinberg's cubic equation to $L(u)$ produces a plane cubic depending on u , and these plane cubics form the associated elliptic fibration. The corresponding satellite cubic is

$$\begin{aligned}
(47) \quad \tilde{\mathcal{F}} = & x_0^2 z + x_1^2 x_2 + (x_2^2 + a_2 x_2 y_2 + b_2 y_2^2)y_2 \\
& + (ax_1 + by_1)y_1 z + (f_2 x_2 + g_2 y_2)y_2 z \\
& + (a_1 x_1 + f_1 x_2 + b_1 y_1 + h y_2 + g_1 z)z^2.
\end{aligned}$$

These equations are given in [66, Equations (17) and (18)]. The degree-eight surface $Y_V \subset \mathcal{S}_{\text{sc}}$ is the common residual component of the two cubic sections. Equivalently, it is defined inside the scroll by

$$(48) \quad \mathcal{F} = \tilde{\mathcal{F}} = 0.$$

The minimal resolution of Y_V is a K3 surface [66, Equation (20)]. The cubic $\mathcal{F}$ does not cut out the surface Y_V alone. Indeed, since the scroll $\mathcal{S}_{\text{sc}}$ has degree three, a cubic hypersurface section has degree nine, whereas Y_V has degree eight. Write v_1 for the point $u = 0$ and v_2 for the point $u = \infty$ on the ruling base. Vinberg shows that

$$(49) \quad V_{\mathcal{S}_{\text{sc}}}(\mathcal{F}) = Y_V \cup L(v_2),$$

where $L(v_2) \cong \mathbb{P}^2$ is one ruling plane of the scroll. The satellite cubic satisfies

$$(50) \quad V_{\mathcal{S}_{\text{sc}}}(\tilde{\mathcal{F}}) = Y_V \cup L(v_1),$$

where $L(v_1)$ is a different ruling plane. Consequently,

$$(51) \quad Y_V = V_{\mathcal{S}_{\text{sc}}}(\mathcal{F}, \tilde{\mathcal{F}}).$$

Although two cubic equations occur here, they do not impose two independent conditions on the threefold scroll. In the function field of $\mathcal{S}_{\text{sc}}$ they are related by

$$(52) \quad \tilde{\mathcal{F}} = \frac{x_2}{x_1} \mathcal{F}, \quad \frac{x_2}{x_1} = \frac{y_2}{z} = \frac{z}{y_1},$$

where the second set of identities follows from the scroll relations. On the fiber

$$x_2 = u x_1, \quad y_2 = u^2 y_1, \quad z = u y_1,$$

one therefore has

$$(53) \quad \tilde{\mathcal{F}} \Big|_{L(u)} = u \mathcal{F} \Big|_{L(u)}.$$

Thus, on a general ruling, $\mathcal{F}$ and $\tilde{\mathcal{F}}$ define the same plane cubic, and it suffices to use $\mathcal{F}$ in order to derive the Weierstrass equation. The satellite cubic becomes essential at the distinguished fiber $u = \infty$, where $\mathcal{F}$ vanishes identically. Its restriction to that fiber is

$$(54) \quad \tilde{\mathcal{F}} \Big|_{L(v_2)} = y_2 (x_2^2 + a_2 x_2 y_2 + b_2 y_2^2),$$

and hence records the three distinguished lines used in the subsequent neighbor construction.

4.3. The standard and alternate fibrations. The ruling of the scroll is parametrized, on an affine chart of the base, by u . The fiber over u is obtained by setting

$$(55) \quad x_2 = ux_1, \quad y_2 = u^2y_1, \quad z = uy_1.$$

Substitution of (55) into (46) gives

$$(56) \quad \mathcal{F} = x_0^2y_1 + x_1^3 + u^3x_1^2y_1 + \phi_2(u)x_1y_1^2 + \phi_3(u)y_1^3,$$

where

$$(57) \quad \begin{aligned} \phi_2(u) &= a_2u^4 + f_2u^3 + f_1u^2 + a_1u + a, \\ \phi_3(u) &= b_2u^5 + g_2u^4 + hu^3 + g_1u^2 + b_1u + b. \end{aligned}$$

This is Vinberg's fiberwise Weierstrass form in [66, Equations (23)–(25)].

We have the following:

Proposition 4.1. *Vinberg's scroll model $Y_V^{\min}$ is birational to the Weierstrass model (62). The explicit relation between Vinberg's coefficients and those of the Weierstrass model is given in Section D. In particular, one has $H_8 = -\frac{1}{4}(a_2^2 - 4b_2)$.*

Proof. On the affine chart $y_1 \neq 0$, put

$$(58) \quad x_V = \frac{x_1}{y_1}, \quad y_V = i\frac{x_0}{y_1}.$$

Equation (56) becomes

$$(59) \quad y_V^2 = x_V^3 + u^3x_V^2 + \phi_2(u)x_V + \phi_3(u).$$

Define

$$(60) \quad r(u) = -\frac{a_2u + f_2}{2}, \quad x_V = x + r(u).$$

We have

$$(61) \quad \begin{aligned} &x_V^3 + u^3x_V^2 + \phi_2x_V + \phi_3 \\ &= x^3 + (u^3 + 3r)x^2 + (\phi_2 + 2u^3r + 3r^2)x + (\phi_3 + \phi_2r + u^3r^2 + r^3). \end{aligned}$$

For the particular choice in (60), the terms u^4 and u^3 in the coefficient of x cancel. Equation (59) takes the normalized form

$$(62) \quad \begin{aligned} X_{F,G,H} : \quad y^2 &= x^3 + (u^3 + F_4u + F_6)x^2 \\ &+ (G_8u^2 + G_{10}u + G_{12})x \\ &+ H_8u^5 + H_{10}u^4 + H_{12}u^3 + H_{14}u^2 + H_{16}u + H_{18}. \end{aligned}$$

This is the standard Weierstrass equation (62). In particular, the multiset of weights of the eleven parameters is

$$4, 6, 8, 8, 10, 10, 12, 12, 14, 16, 18,$$

in agreement with Vinberg's free algebra of automorphic forms. Together with the following rational coordinate map, they identify the two birational surface models. On the open charts used above, the map from the Weierstrass model (62) to the scroll is

$$(63) \quad \begin{aligned} (u, x, y) &\dashrightarrow [x_0 : x_1 : x_2 : y_1 : y_2 : z] \\ &= [-iy : x + r(u) : u(x + r(u)) : 1 : u^2 : u]. \end{aligned}$$

The inverse map is

$$(64) \quad u = \frac{z}{y_1}, \quad x = \frac{x_1}{y_1} - r(u), \quad y = i \frac{x_0}{y_1}.$$

The maps are birational on the finite- u chart. Since smooth K3 surfaces that are birational are isomorphic, the two minimal resolutions are the same K3 surface. $\square$

The second isometric presentation

$$U \oplus D_8(-1) \cong U(2) \oplus E_8(-1)$$

is also visible in Vinberg's distinguished scroll fiber. Recall that restricting the satellite cubic to the fiber at the distinguished boundary point gives the three lines

$$(65) \quad y_2(x_2^2 + a_2x_2y_2 + b_2y_2^2) = 0.$$

The two lines belonging to the quadratic factor are determined by the roots of

$$(66) \quad \rho^2 + a_2\rho + b_2 = 0.$$

If we introduce $\delta^2 = a_2^2 - 4b_2$, the two roots of Equation (66) are $\varrho_{\pm} = \frac{-a_2 \pm \delta}{2}$. Using (158), we find $a_2^2 - 4b_2 = -4H_8$. Thus, after adjoining a choice of square root, we may set

$$(67) \quad \delta = 2H_4, \quad H_8 = -H_4^2.$$

Remark 4.2. At a fixed point in the moduli space with $H_8 \neq 0$, either square root of the discriminant of (66) may be chosen. Adjoining $\delta = 2H_4$ to the parameter space defines the double cover

$$(68) \quad \delta^2 = -4H_8.$$

The nontrivial deck transformation $\delta \mapsto -\delta$ interchanges the two roots, hence the two lines in (65). Thus H_8 is defined on the full coefficient quotient, whereas $H_4 = \delta/2$ is defined on the twofold cover that selects one line. For the period lattice $T = U^{\oplus 2} \oplus D_8(-1)$, the quotient $O^+(T)/\tilde{O}^+(T)$ is cyclic of order two and its nontrivial element interchanges the two isotropic discriminant classes; the form denoted G_4 in [10, Section 5] is H_4 in our normalization. Its branch divisor is

$$(69) \quad H_8 = 0 \quad \Leftrightarrow \quad a_2^2 = 4b_2.$$

Along this divisor the two lines in the quadratic factor of (65) merge. This is Vinberg's codimension-one specialization in [66, Equation (22)]. The resolved K3 surface generally remains smooth but acquires an additional algebraic class; the polarization enhances from $U \oplus D_8(-1)$ to one containing $U \oplus D_9(-1)$.

The choice of square root in a 2-neighbor construction corresponds to the choice of one of the two lines in the quadratic factor of (65). We have the following:

Proposition 4.3. *A very general K3 surface X with $\text{NS}(X) = U \oplus D_8(-1)$ admits a birational model as a double cover of $\mathbb{P}_{(s)}^1 \times \mathbb{P}_{(u)}^1$, given on an affine chart by*

$$(70) \quad w^2 = u^4 A_0(s) + u^3 A_1(s) + u^2 A_2(s) + u A_3(s) + A_4(s),$$

with

$$\begin{aligned}
(71) \quad & A_0(s) = 2H_4s + H_{10}, \\
& A_1(s) = s^2 + H_4^3 + F_4H_4^2 + H_4G_8 + H_{12}, \\
& A_2(s) = (3H_4^2 + 2F_4H_4 + G_8)s + F_6H_4^2 + H_4G_{10} + H_{14}, \\
& A_3(s) = (3H_4 + F_4)s^2 + (2F_6H_4 + G_{10})s + H_4G_{12} + H_{16}, \\
& A_4(s) = s^3 + F_6s^2 + G_{12}s + H_{18}.
\end{aligned}$$

Proof. The neighbor substitution in the normalized Weierstrass coordinate is

$$(72) \quad x = s + H_4u.$$

Combining (72) with (60) gives a direct expression in Vinberg's fiber coordinate, namely

$$(73) \quad x_V = s - \frac{f_2}{2} + \varrho_+ u = s - \frac{f_2}{2} + \frac{-a_2 + \delta}{2} u.$$

Replacing δ by $-\delta$ replaces ϱ_+ by ϱ_- and selects the other line. Substitute (72) and $H_8 = -H_4^2$ into (62). The only terms of degree five in u are

$$u^3(H_4u)^2 + H_8u^5 = (H_4^2 + H_8)u^5 = 0.$$

After renaming the left-hand coordinate w , the equation is the quartic (70). In particular,

$$(74) \quad \deg_s(A_4, A_3, A_2, A_1, A_0) = (3, 2, 1, 2, 1).$$

The absence of a section for the very general member follows from the same discriminant argument as in Appendix B: a section would give a full-rank sublattice $U \oplus E_8(-1)$ of discriminant one inside $\text{NS}(X)$ of discriminant four, which is impossible. Since the old fiber is a bisection, the new fiber class has divisibility two. $\square$

Equation (70) realizes the $U(2) \oplus E_8(-1)$ presentation of the same K3 surface $X_{F,G,H}$ as a genus-one fibration over $\mathbb{P}_{(s)}^1$. The relative Jacobian of the index-two genus-one fibration (70) has the Weierstrass equation

$$(75) \quad J_{F,G,H} : \quad \eta^2 = \xi^3 - \frac{1}{3}\mathcal{I}(s)\xi - \frac{1}{27}\mathcal{J}(s),$$

where the classical binary-quartic invariants are

$$\begin{aligned}
(76) \quad & \mathcal{I} = 12A_0A_4 - 3A_1A_3 + A_2^2, \\
& \mathcal{J} = 72A_0A_2A_4 + 9A_1A_2A_3 - 27A_0A_3^2 - 27A_1^2A_4 - 2A_2^3,
\end{aligned}$$

see [18] for these conventions.

For very general coefficients, we choose markings so that

$$(77) \quad T(X_{F,G,H}) \cong T_D := U^{\oplus 2} \oplus D_8(-1), \quad T(J_{F,G,H}) \cong T_E := U^{\oplus 2} \oplus E_8(-1),$$

Ogg-Shafarevich theory associates to the genus-one fibration its Brauer class α on the relative Jacobian. Its order is the multisection index, which is two here. Thus $\alpha \in \text{Br}(J_{F,G,H})[2]$, and it identifies the transcendental lattice of the torsor via

$$T_D \cong \ker(\alpha : T_E \rightarrow \mathbb{Z}/2\mathbb{Z})$$

as an index-two Hodge sublattice; see [42, Lemma 4.4 and Equation (5.1)]. In particular, $T_D \otimes \mathbb{Q} = T_E \otimes \mathbb{Q} =: V$, and the two period lines agree in the type-IV domain $\mathcal{D}(V)$.

4.4. Relation with the Hashimoto–Ueda family. Equation (75) is a Weierstrass model for the family of $U \oplus E_8(-1)$ -polarized K3 surfaces. Such surfaces were also studied by Hashimoto and Ueda in [30]. Their normal form is

$$(78) \quad X_{\text{HU}} : \quad y^2 = x^3 + (C_4 t^4 + C_{10} t^3 + C_{16} t^2 + C_{22} t + C_{28})x \\ + t^7 + C_{12} t^5 + C_{18} t^4 + C_{24} t^3 + C_{30} t^2 + C_{36} t + C_{42}.$$

They prove that the scalar-valued ring for $\Gamma := \text{O}^+(U^{\oplus 2} \oplus E_8(-1))$ is

$$(79) \quad M_*(\Gamma) = \mathbb{C}[C_4, C_{10}, C_{12}, C_{16}, C_{18}, C_{22}, C_{24}, C_{28}, C_{30}, C_{36}, C_{42}],$$

and that the full ring with characters is obtained by adjoining a determinant-character generator Φ_{252} satisfying one relation of weight 504; namely $\Phi_{252}^2 = \Psi_{504}$. We discuss the details in Section C.

Let Γ_D denote the arithmetic group for the Vinberg D_8 -family, and let χ_4 be the character of the weight-four form H_4 . We work on the character cover corresponding to

$$\Gamma_D^{H_4} := \ker(\chi_4).$$

On this cover the entries of

$$(80) \quad \mathbf{D} = (F_4, H_4, F_6, G_8, G_{10}, H_{10}, G_{12}, H_{12}, H_{14}, H_{16}, H_{18})$$

are scalar-valued modular forms. Their weights are

$$\mathbf{d} = (4, 4, 6, 8, 10, 10, 12, 12, 14, 16, 18),$$

and hence $\sum_i d_i = 114$. Similarly, set

$$(81) \quad \mathbf{C} = (C_4, C_{10}, C_{12}, C_{16}, C_{18}, C_{22}, C_{24}, C_{28}, C_{30}, C_{36}, C_{42}),$$

with weight vector

$$\mathbf{c} = (4, 10, 12, 16, 18, 22, 24, 28, 30, 36, 42),$$

so that $\sum_j c_j = 242$. Since the relevant type-IV domains have dimension ten, Proposition C.1 gives modular Jacobians of weights 124 and 252. We write

$$(82) \quad \phi_{124} := J(\mathbf{D}), \quad J(\mathbf{C}) \doteq \Phi_{252},$$

where $\doteq$ denotes equality up to a nonzero constant. Thus ϕ_{124} has determinant character for $\Gamma_D^{H_4}$; viewed on the full group Γ_D , its character is $\chi_4 \det$. The second relation in (82) follows from Theorem C.2 and the normalization $\Phi_{252}^2 = \Psi_{504}$; see [30, 67].

Lemma 4.4. *The inclusion of lattices induces an inclusion of arithmetic groups*

$$\Gamma_D^{H_4} = \tilde{\text{O}}^+(T_D) \subseteq \Gamma_E := \text{O}^+(T_E)$$

and hence a finite morphism

$$\Gamma_D^{H_4} \backslash \mathcal{D}(V) \longrightarrow \Gamma_E \backslash \mathcal{D}(V).$$

In a single rational system of tube-domain coordinates for $V = T_D \otimes \mathbb{Q} = T_E \otimes \mathbb{Q}$, this morphism is the identity in period coordinates, and the determinant character of Γ_E restricts to the determinant character of $\Gamma_D^{H_4}$.

Proof. The overlattice $T_E \supset T_D$ corresponds to the isotropic subgroup $K := T_E/T_D \subset A_{T_D}$ of order two. By Remark 4.2, $\Gamma_D^{H_4} = \tilde{O}^+(T_D)$. Every element of this discriminant kernel fixes K and therefore extends to an isometry of T_E [49]. This proves the inclusion of groups; it has finite index because both groups are arithmetic in $O^+(V)$. Since the lattice inclusion is a Hodge inclusion, it preserves the period line. The final statements follow by choosing one rational tube-domain basis for V ; the determinant character in both cases is the determinant of the action on this common quadratic space. $\square$

Proposition 4.5. *The relative Jacobian of the genus-one fibration (70) determines a weighted-homogeneous polynomial map of affine coefficient cones*

$$\Theta: \mathbf{D} \mapsto \mathbf{C}(\mathbf{D}),$$

given by weighted-homogeneous polynomials $C_k = C_k(F, G, H)$. The ordinary coefficient Jacobian

$$(83) \quad J_\Theta := \text{Jac}(\Theta) = \det \left(\frac{\partial C_j}{\partial D_i} \right)_{1 \leq i, j \leq 11}$$

is weighted homogeneous of weight 128. In the compatible tube-domain coordinates of Lemma 4.4, one has

$$(84) \quad J(\mathbf{C}(\mathbf{D})) = J_\Theta J(\mathbf{D}) \quad \Leftrightarrow \quad \Theta^* \Phi_{252} \doteq J_\Theta \phi_{124}.$$

Proof. Assign weights

$$\text{wt}(u) = 2, \quad \text{wt}(s) = 6, \quad \text{wt}(w) = 9.$$

The formulas in (71) show that $A_i(s)$ is weighted homogeneous of weight $10 + 2i$. It follows from (76) that

$$\text{wt}(\mathcal{I}) = 28, \quad \text{wt}(\mathcal{J}) = 42.$$

The degree pattern $\deg_s(A_4, A_3, A_2, A_1, A_0) = (3, 2, 1, 2, 1)$ also gives

$$\deg_s \mathcal{I} = 4, \quad \deg_s \mathcal{J} = 7$$

for general coefficients. More precisely,

$$A_1(s) = s^2 + O(1), \quad A_4(s) = s^3 + F_6 s^2 + O(s).$$

Among the summands in $\mathcal{J}$, only $-27A_1^2 A_4$ can have degree at least six. Therefore

$$(85) \quad -\frac{1}{27} \mathcal{J}(s) = s^7 + F_6 s^6 + O(s^5).$$

The weighted translation $s = t - F_6/7$ eliminates the coefficient of t^6 . Define

$$(86) \quad f_{\text{HU}}(t) := -\frac{1}{3} \mathcal{I} \left(t - \frac{F_6}{7} \right), \quad g_{\text{HU}}(t) := -\frac{1}{27} \mathcal{J} \left(t - \frac{F_6}{7} \right).$$

Then $\deg_t f_{\text{HU}} = 4$, $\deg_t g_{\text{HU}} = 7$, the leading coefficient of g_{HU} is one, and its t^6 -coefficient vanishes. Comparison with (78) yields every C_k as a polynomial in the entries of $\mathbf{D}$, and the weight computation above shows that it is weighted homogeneous of weight k .

It remains to verify the modular-Jacobian identity. Write D_i and C_j for the entries of $\mathbf{D}$ and $\mathbf{C}$, let $d_i = \text{wt}(D_i)$ and $c_j = \text{wt}(C_j)$, and put

$$P_{ij} := \frac{\partial C_j}{\partial D_i}.$$

Weighted Euler homogeneity gives

$$c_j C_j = \sum_{i=1}^{11} d_i D_i P_{ij}.$$

For every tube-domain coordinate z_ν , the ordinary chain rule gives

$$\partial_{z_\nu} C_j = \sum_{i=1}^{11} (\partial_{z_\nu} D_i) P_{ij}.$$

If $M_{\mathbf{D}}$ and $M_{\mathbf{C}}$ denote the matrices occurring in the modular Jacobians $J(\mathbf{D})$ and $J(\mathbf{C}(\mathbf{D}))$, these two identities say precisely that

$$M_{\mathbf{C}} = M_{\mathbf{D}} P.$$

Taking determinants proves (84). Moreover, every term of $\det P$ has weight

$$\sum_{j=1}^{11} c_j - \sum_{i=1}^{11} d_i = 242 - 114 = 128,$$

so J_{Θ} has weight 128. Finally, substituting $J(\mathbf{D}) = \phi_{124}$ and $J(\mathbf{C}) \doteq \Phi_{252}$ proves (84). $\square$

4.4.1. *The A_7 -enhancement divisor in the $U \oplus D_8(-1)$ family.* The general $U \oplus D_8(-1)$ -polarized K3 surface was given by (62). We wish to characterize the sublocus for which the transcendental lattice specializes from $U^{\oplus 2} \oplus D_8(-1)$ to $U^{\oplus 2} \oplus A_7(-1)$. An explicit model for the latter family is

$$(87) \quad y^2 = x^3 + (u^3 + u f_4 + f_6) x^2 + (u^4 g_4 + u^3 g_6 + u^2 g_8 + u g_{10} + g_{12}) x + (u^2 h_5 + u h_7 + h_9)^2.$$

There are two $\tilde{O}^+$ -inequivalent embeddings $A_7 \hookrightarrow D_8$; their union is the Heegner divisor of the weight-128 Borcherds product Φ_{128} [10]. In [10], the coefficient specialization is made explicit. Let h_5 be a variable independent of the coefficients of (62). On the chart $H_4 \neq 0$, apply the substitution

$$(88) \quad x \mapsto x \pm \left(H_4 u + \frac{h_5^2 - H_{10}}{2H_4} \right).$$

Then Vinberg's D_8 model coincides with the A_7 model (87) if and only if the coefficients F_k, G_k, H_k and f_k, g_k, h_k satisfy the relations in [10, Equation (5.2)].

We observe that after the substitution (88), Equation (62) takes the form of a long Weierstrass model whose constant term is a quartic polynomial in u ,

$$(89) \quad Q(u) = b_0 u^4 + b_1 u^3 + b_2 u^2 + b_3 u + b_4.$$

The A_7 family is characterized by the condition $Q(u) = (h_5 u^2 + h_7 u + h_9)^2$. The Zariski closure of the locus of quartic squares is the image of

$$(90) \quad (r_0, r_1, r_2) \longmapsto (b_0, b_1, b_2, b_3, b_4) = (r_0^2, 2r_0 r_1, r_1^2 + 2r_0 r_2, 2r_1 r_2, r_2^2).$$

Eliminating r_0, r_1, r_2 from (90) shows that its homogeneous ideal is generated by the following seven cubics:

$$\begin{aligned}
& b_3^3 - 4b_2b_3b_4 + 8b_1b_4^2, \\
& b_2b_3^2 - 4b_2^2b_4 + 2b_1b_3b_4 + 16b_0b_4^2, \\
& b_1b_3^2 - 4b_1b_2b_4 + 8b_0b_3b_4, \\
& b_1^2b_4 - b_0b_3^2, \\
& b_1^2b_3 - 4b_0b_2b_3 + 8b_0b_1b_4, \\
& b_1^2b_2 - 4b_0b_2^2 + 2b_0b_1b_3 + 16b_0^2b_4, \\
& b_1^3 - 4b_0b_1b_2 + 8b_0^2b_3.
\end{aligned}
\tag{91}$$

5. EXCEPTIONAL INDEX-TWO JACOBIAN TRANSITIONS OF TYPES E_7 AND E_6

The transition from the D_8 family to the Hashimoto–Ueda E_8 family is based on the lattice isomorphism $U \oplus D_8(-1) \cong U(2) \oplus E_8(-1)$. Its essential arithmetic input is the index-two inclusion $D_8 \subset E_8$, together with the fact that the new fibration is a genus-one torsor of index two. There are two further exceptional inclusions of the same kind:

$$A_7 \subset E_7, \quad A_5 \oplus A_1 \subset E_6, \quad [E_7 : A_7] = [E_6 : A_5 \oplus A_1] = 2.
\tag{92}$$

The determinants on the two sides of each inclusion differ by a factor of four. The corresponding transcendental lattices are

$$\begin{aligned}
T_{A_7} &= U^{\oplus 2} \oplus A_7(-1) \subset T_7 := U^{\oplus 2} \oplus E_7(-1), \\
T_{A_5+A_1} &= U^{\oplus 2} \oplus A_5(-1) \oplus A_1(-1) \subset T_6 := U^{\oplus 2} \oplus E_6(-1).
\end{aligned}
\tag{93}$$

Both are index-two inclusions of even lattices. Thus, exactly as in Equation (77), a genus-one torsor with transcendental lattice on the left can have a relative Jacobian with transcendental lattice on the right. The corresponding Néron–Severi lattices are

$$S_7 = U \oplus E_8(-1) \oplus A_1(-1), \quad S_6 = U \oplus E_8(-1) \oplus A_2(-1).
\tag{94}$$

Wang and Williams proved that the relevant rings of modular forms are free [69, Theorem 5.9]; see also [67]. Theorems 5.1 and 5.2 below identify the geometric coefficients with generating systems of these rings. In this notation, the rings are

$$M_*(O^+(T_7)) = \mathbb{C}[P_4, P_6, P_{10}, P_{12}, P_{14}, P_{16}, P_{18}, P_{22}, P_{24}, P_{30}],
\tag{95}$$

and

$$M_*(\widetilde{O}^+(T_6)) = \mathbb{C}[Q_4, Q_6, Q_7, Q_{10}, Q_{12}, Q_{15}, Q_{16}, Q_{18}, Q_{24}].
\tag{96}$$

The subscripts give the modular weights. In the E_7 case all ten generators can be chosen to be orthogonal Eisenstein series. In the E_6 case the odd-weight generators can be chosen as two additive lifts of holomorphic $W(E_6)$ -invariant Jacobi forms of weights 7 and 15 and index one [69, Section 5.3]; see [55, 70] for background on the associated weak Jacobi forms.

5.1. The relative Jacobian of the A_7 family. We start with the A_7 model in Equation (87). We now regard x as the base coordinate and u as the coordinate on the quartic fiber. The same surface is then a genus-one fibration

$$(97) \quad w^2 = A_0(x)u^4 + A_1(x)u^3 + A_2(x)u^2 + A_3(x)u + A_4(x),$$

where

$$(98) \quad \begin{aligned} A_0(x) &= g_4x + h_5^2, \\ A_1(x) &= x^2 + g_6x + 2h_5h_7, \\ A_2(x) &= g_8x + h_7^2 + 2h_5h_9, \\ A_3(x) &= f_4x^2 + g_{10}x + 2h_7h_9, \\ A_4(x) &= x^3 + f_6x^2 + g_{12}x + h_9^2. \end{aligned}$$

Let $\mathcal{I}(x)$ and $\mathcal{J}(x)$ be the binary-quartic invariants in Equation (76), and put

$$(99) \quad D = h_7^2 - 4h_5h_9.$$

At $x = 0$ the quartic in (97) is $(h_5u^2 + h_7u + h_9)^2$, and

$$(100) \quad \mathcal{I}(0) = D^2, \quad \mathcal{J}(0) = -2D^3.$$

The relative Jacobian is in short Weierstrass form

$$\eta^2 = \xi^3 - \frac{1}{3}\mathcal{I}(x)\xi - \frac{1}{27}\mathcal{J}(x).$$

After the translation $\xi = T + D/3$, it becomes

$$(101) \quad \eta^2 = T^3 + DT^2 + B_7(x)T + C_7(x),$$

where

$$(102) \quad B_7(x) = \frac{D^2 - \mathcal{I}(x)}{3}, \quad C_7(x) = \frac{D^3 - 3D\mathcal{I}(x) - \mathcal{J}(x)}{27}.$$

We obtain

$$(103) \quad \begin{aligned} B_7(x) &= P_4x^4 + P_{10}x^3 + P_{16}x^2 + P_{22}x, \\ C_7(x) &= x^7 + P_6x^6 + P_{12}x^5 + P_{18}x^4 + P_{24}x^3 + P_{30}x^2, \end{aligned} \quad P_{14} = D.$$

Thus, the relative Jacobian over $\mathbb{P}_{(x)}^1$ has the following normal form:

$$(104) \quad \begin{aligned} \mathcal{X}_7 : \quad \eta^2 &= T^3 + P_{14}T^2 + (P_4x^4 + P_{10}x^3 + P_{16}x^2 + P_{22}x)T \\ &\quad + x^7 + P_6x^6 + P_{12}x^5 + P_{18}x^4 + P_{24}x^3 + P_{30}x^2. \end{aligned}$$

We have the following:

Theorem 5.1. *For very general coefficients, the minimal resolution of (104) is an S_7 -polarized K3 surface with*

$$\mathrm{NS}(\mathcal{X}_7) \cong U \oplus E_8(-1) \oplus A_1(-1), \quad T(\mathcal{X}_7) \cong U^{\oplus 2} \oplus E_7(-1).$$

Equation (104) has singular fibers $\mathrm{II}^* + \mathrm{I}_2 + 12\mathrm{I}_1$ and trivial Mordell–Weil group. With the grading $\mathrm{wt}(x) = 6$, $\mathrm{wt}(T) = 14$, $\mathrm{wt}(\eta) = 21$, the ten coefficients have weights

$$4, 6, 10, 12, 14, 16, 18, 22, 24, 30.$$

After normalization, they form a free generating system for the ring in Equation (95).

Proof. At $x = \infty$, set $s = 1/x$ and make the minimal change of variables $T = s^{-4}\tilde{T}$, $\eta = s^{-6}\tilde{\eta}$. The resulting vanishing orders are those of a fiber of type II^* . At $x = 0$, the cubic on the right side of (104) is $T^2(T + P_{14})$. The cubic discriminant has expansion

$$(105) \quad \text{disc}_T(T^3 + P_{14}T^2 + B_7(x)T + C_7(x)) = P_{14}^2(P_{22}^2 - 4P_{14}P_{30})x^2 + O(x^3).$$

It follows that the fiber over zero is generically of type I_2 . The remaining Euler number is twelve, and the remaining singular fibers are nodal for a very general coefficient vector. The fiber, zero section, and components of the II^* and I_2 fibers generate $U \oplus E_8(-1) \oplus A_1(-1)$. The polarizing lattice is generated by the classes of the fiber, zero section, and nonidentity components of the reducible fibers. An isomorphism preserving this polarization fixes 0 and ∞ as the base points of these fibers. Between two Weierstrass models it has the form

$$x \mapsto ax, \quad T \mapsto b^2T + r(x), \quad \eta \mapsto b^3\eta,$$

with $a, b \neq 0$. The constant coefficient of T^2 forces r to be constant, and the node at $(x, T, \eta) = (0, 0, 0)$ forces $r = 0$. The monic x^7 term gives $b^6 = a^7$, and we obtain

$$(x, T, \eta) \mapsto (\lambda^6x, \lambda^{14}T, \lambda^{21}\eta).$$

Thus the coefficient space modulo this scaling is generically finite over the moduli space of the indicated lattice-polarized surfaces and has dimension nine. The very general member consequently has Picard rank eleven. Since the constructed trivial lattice has discriminant two, it admits no proper integral overlattice; hence it is the full Néron–Severi lattice and the Mordell–Weil group is trivial. Its orthogonal complement in the K3 lattice is T_7 .

The old elliptic fiber supplies a bisection of (97). A section is excluded by the discriminants: the torsor has Néron–Severi discriminant eight, whereas the displayed Jacobian trivial lattice has discriminant two. Ogg–Shafarevich theory therefore identifies T_{A_7} with the kernel of a nonzero two-torsion Brauer class on the relative Jacobian, as in [42, Lemma 4.4 and Equation (5.1)]. The unique even index-two overlattice is T_7 . Thus the relative-Jacobian period map is generically finite.

We next establish holomorphy on the whole period domain. By [10, Theorem 1.1(ii) and Section 4.7], the ten A_7 coefficient forms are holomorphic on the affine period cone of T_{A_7} . The index-two inclusion identifies this cone with that of T_7 . Equations (99) and (102), or the expanded formulas in Appendix E, express every P_k as a homogeneous polynomial in those forms. Hence every P_k is holomorphic on the entire cone, including the divisors omitted from the generic geometric description. To check descent to the larger arithmetic group, fix the primitive embedding $S_7 \hookrightarrow \Lambda_{\text{K3}}$. Its pointwise stabilizer induces $\tilde{\text{O}}^+(T_7)$ on the orthogonal complement; this is $\text{O}^+(T_7)$ because the discriminant group has order two [49, 14]. On the generic locus, the lattice-polarized Torelli theorem and the uniqueness of the Weierstrass normal form identify equivalent periods with the same normal form up to scaling. The holomorphic two-form $\frac{dx \wedge dT}{\eta}$ scales by λ^{-1} . The canonical identification of invariant differentials on a genus-one curve and its Jacobian identifies the holomorphic two-form with the torsor differential up to a fixed nonzero numerical factor; see [18]. Thus on the affine period cone, the coefficient of weight k also satisfies $P_k(c\omega) = c^{-k}P_k(\omega)$ and is invariant under $\text{O}^+(T_7)$. The invariance then holds everywhere by the identity theorem. Interior holomorphy has already been proved, and Koecher’s principle gives holomorphy at the cusps. Hence $P_k \in M_k(\text{O}^+(T_7))$.

The nine-dimensional moduli image together with the nontrivial Hodge scaling proves algebraic independence of the ten coefficients. Equivalently, differentiating the formulas in Appendix E in the input order $(f_4, f_6, g_4, g_6, g_8, g_{10}, g_{12}, h_5, h_7, h_9)$ gives rank ten at $(1, 2, \dots, 10)$. Their weights agree with the free generator weights in (95). The resulting polynomial subalgebra therefore has the same Hilbert series as the full ring, and equality holds in every weight. $\square$

5.2. The E_6 Heegner divisor and the two odd generators. Equation (105) gives the enhancement from A_1 to A_2 . Define the weight-44 polynomial

$$(106) \quad \mathcal{B}_{44} := P_{22}^2 - 4P_{14}P_{30}.$$

On the open set $P_{14} \neq 0$, the condition $\mathcal{B}_{44} = 0$ generically enhances the fiber at $x = 0$ from I_2 to I_3 . The resulting general K3 surface has Néron–Severi and transcendental lattices S_6 and T_6 , respectively. On the graded coefficient cone for the discriminant kernel, the rank-one condition in (106) has the parametrization

$$(107) \quad P_{14} = Q_7^2, \quad P_{22} = 2Q_7Q_{15}, \quad P_{30} = Q_{15}^2.$$

Restricting the common even generators by $P_k = Q_k$ gives the equation

$$(108) \quad \begin{aligned} \mathcal{X}_6 : \quad \eta^2 = T^3 + Q_7^2 T^2 + (Q_4 x^4 + Q_{10} x^3 + Q_{16} x^2 + 2Q_7 Q_{15} x) T \\ + x^7 + Q_6 x^6 + Q_{12} x^5 + Q_{18} x^4 + Q_{24} x^3 + Q_{15}^2 x^2. \end{aligned}$$

We will write $B_6(x)$ and $C_6(x)$ for the coefficient of T and the constant term, respectively, on the right side of (108). Equivalently, we can shift $\eta = \tilde{\eta} + Q_7 T + Q_{15} x$ to obtain

$$(109) \quad \begin{aligned} \mathcal{X}_6 : \quad \tilde{\eta}^2 + 2(Q_{15} x + Q_7 T) \tilde{\eta} = T^3 + (Q_4 x^2 + Q_{10} x + Q_{16}) x^2 T \\ + (x^4 + Q_6 x^3 + Q_{12} x^2 + Q_{18} x + Q_{24}) x^3. \end{aligned}$$

This shows that there is a suitable normal form where each generator appears by itself.

The specialization from the A_7 equation to the $A_5 \oplus A_1$ equation can be made explicit. We set

$$L(u) = h_5 u + h_7', \quad G(u) = g_4 u^3 + g_6' u^2 + g_8' u + g_{10}',$$

and assume that the last term and the coefficient of x in Equation (87) are

$$L(u)^2(u - a_2)^2, \quad G(u)(u - a_2),$$

respectively, where a_2 is the position of the additional I_2 -fiber in $\mathbb{P}_{(u)}^1$. As an A_7 specialization, this means

$$(110) \quad \begin{aligned} (h_5, h_7, h_9)_{A_7} &= (h_5, h_7' - a_2 h_5, -a_2 h_7'), \\ (g_4, g_6, g_8, g_{10}, g_{12})_{A_7} &= (g_4, g_6' - a_2 g_4, g_8' - a_2 g_6', g_{10}' - a_2 g_8', -a_2 g_{10}'). \end{aligned}$$

In particular, the discriminant in (99) becomes

$$(111) \quad D = (h_7' + a_2 h_5)^2.$$

Set

$$(112) \quad Q_7 = L(a_2) = h_7' + a_2 h_5, \quad Q_{15} = h_5 G(a_2) - \frac{1}{3} Q_7 G'(a_2).$$

A direct substitution into the polarized binary-quartic invariants then gives

$$(113) \quad P_{14} = Q_7^2, \quad P_{22} = 2Q_7 Q_{15}, \quad P_{30} = Q_{15}^2.$$

In particular, the square root of the quadratic discriminant is the weight-seven generator, while the tangent coefficient in the exchanged quartic is the weight-fifteen generator.

Theorem 5.2. *For very general coefficients, the minimal resolution of (108) is an S_6 -polarized K3 surface with*

$$\mathrm{NS}(\mathcal{X}_6) \cong U \oplus E_8(-1) \oplus A_2(-1), \quad T(\mathcal{X}_6) \cong U^{\oplus 2} \oplus E_6(-1).$$

Equation (108) has singular fibers $\mathrm{II}^ + \mathrm{I}_3 + 11\mathrm{I}_1$ and trivial Mordell–Weil group. The nine quantities*

$$Q_4, Q_6, Q_7, Q_{10}, Q_{12}, Q_{15}, Q_{16}, Q_{18}, Q_{24}$$

form a free generating system for $M_(\tilde{\mathcal{O}}^+(T_6))$ after normalization.*

Proof. The fiber at infinity is again of type II^* . At zero the terms of lowest order in (T, x) are

$$Q_7^2 T^2 + 2Q_7 Q_{15} x T + Q_{15}^2 x^2 = (Q_7 T + Q_{15} x)^2.$$

More precisely, put

$$(114) \quad \mathcal{B}_{45} = Q_{15}^3 + Q_7^2 Q_{15} Q_{16} - Q_7^3 Q_{24}.$$

Then

$$(115) \quad \mathrm{disc}_T(T^3 + Q_7^2 T^2 + B_6(x)T + C_6(x)) = 4Q_7^3 \mathcal{B}_{45} x^3 + O(x^4).$$

Hence the fiber is generically I_3 . The same Euler-number, dimension, and discriminant argument used in Theorem 5.1 gives the asserted fiber configuration, Néron–Severi lattice, and trivial Mordell–Weil group. The old fiber again supplies a bisection; a section is excluded by the corresponding discriminants twelve and three. The unique even index-two overlattice of $T_{A_5+A_1}$ is T_6 , so the period map of the relative-Jacobian construction is generically finite.

For interior holomorphy, use the embedding $T_{A_5+A_1} = T_{A_7} \cap T_6$ and restrict the A_7 coefficient forms to this period subdomain; its affine cone is also that of T_6 . The seven even coefficients are restrictions of the holomorphic P_k . Write

$$h(u) = h_5 u^2 + h_7 u + h_9, \quad g(u) = g_4 u^4 + g_6 u^3 + g_8 u^2 + g_{10} u + g_{12}.$$

On a dense open subset of this subdomain, g and h have the unique common root a_2 . Division of g by h gives a linear remainder $R_1 u + R_0$, with rational expressions in the coefficient forms, and $a_2 = -R_0/R_1$. Thus a_2 , $L = h/(u - a_2)$, and $G = g/(u - a_2)$ have meromorphic coefficients on the cone. In particular, (112) defines single-valued meromorphic functions Q_7 and Q_{15} there. The identities (113) hold meromorphically, so $Q_7^2 = P_{14}|_{T_6}$ and $Q_{15}^2 = P_{30}|_{T_6}$ are holomorphic; neither Q_7 nor Q_{15} can therefore have a pole. Normality of the period cone extends them holomorphically across the remaining subsets of codimension at least two.

For descent, the S_6 -polarization fixes the two nonidentity components of the I_3 fiber. Their ordering distinguishes the two branches $\eta = \pm Q_7 T$ of the nodal cubic at $x = 0$, and hence fixes the simultaneous sign of Q_7, Q_{15} in (107). Choose this ordering to agree with (112) at one generic point. The pointwise stabilizer of S_6 induces exactly $\tilde{\mathcal{O}}^+(T_6)$, whereas the interchange of these components acts nontrivially on its discriminant group. The same normal-form and differential argument in the proof of Theorem 5.1, while also preserving this ordering, shows invariance under the discriminant kernel and homogeneity of weight k for each Q_k . The identity theorem and Koecher’s principle then give $Q_k \in M_k(\tilde{\mathcal{O}}^+(T_6))$ on the whole modular variety.

The coefficient space modulo scaling has dimension eight; hence the nine forms are algebraically independent. An explicit check is that their Jacobian, after (110) and (112), has rank nine at $(1, 2, \dots, 9)$ in the input order $(f_4, f_6, g_4, g'_6, g'_8, g'_{10}, h_5, h'_7, a_2)$. Comparison with the Hilbert series in (96) proves that they freely generate the full ring. $\square$

Equation (108) depends on Q_7 and Q_{15} only through their quadratic monomials. This is because of the difference between the full stabilizer of the E_6 Heegner divisor and its discriminant-kernel cover. In particular, if we let ϵ denote the nontrivial element of $O(A_{T_6}) \cong \mathbb{Z}/2\mathbb{Z}$, then it acts by

$$\epsilon(Q_7, Q_{15}) = (-Q_7, -Q_{15})$$

and fixes the seven even generators. Therefore, we have

$$(116) \quad \frac{M_*(O^+(T_7))}{(\mathcal{B}_{44})} \cong M_*(\tilde{O}^+(T_6))^{\langle \epsilon \rangle},$$

where the homomorphism is given by (107). In particular, the odd generators are not individually visible on the unmarked E_6 divisor; the relative-Jacobian construction naturally produces their simultaneous sign cover on the graded coefficient cone. This does not give a twofold cover of the coarse projective moduli space: the sign change is the weighted scaling $\lambda = -1$, and $-\text{id} \in O^+(T_6)$ acts trivially on the projective period domain.

5.3. Borchers products and modular Jacobians. The two local discriminant calculations identify distinguished Heegner divisors. A primitive vector $v \in T_7$ with

$$v^2 = -6, \quad \text{div}(v) = 2$$

has $v^\perp \cong T_6$. Similarly, a primitive vector $w \in T_6$ with

$$w^2 = -12, \quad \text{div}(w) = 3$$

has $w^\perp \cong U^{\oplus 2} \oplus D_5(-1)$. These are the Heegner orbits associated with the minuscule-weight terms in the weight-zero E_7 and E_6 Weyl-invariant weak Jacobi forms. Their Borchers lifts have weights 44 and 45, respectively [70, 25, 69].

Corollary 5.3. *Up to nonzero constants, the exceptional Borchers products are*

$$(117) \quad \Phi_{44, E_7} \doteq P_{22}^2 - 4P_{14}P_{30}, \quad \Phi_{45, E_6} \doteq Q_{15}^3 + Q_7^2 Q_{15} Q_{16} - Q_7^3 Q_{24}.$$

The first cuts out the E_6 locus inside the E_7 modular variety. On the open set $Q_7 \neq 0$, the second generically raises the fiber at $x = 0$ from I_3 to I_4 ; its general point therefore has

$$\text{NS} \cong U \oplus E_8(-1) \oplus A_3(-1), \quad T \cong U^{\oplus 2} \oplus D_5(-1).$$

Proof. The two theorems identify the coefficient rings with the full modular-form rings. In these polynomial rings both displayed polynomials are irreducible, so each defines a reduced irreducible divisor. Equations (105) and (115) identify its general point with the stated lattice enhancement. The corresponding Heegner divisors are irreducible for the indicated groups, so these are their entire supports. There is no generic divisorial ramification here: neither the reflection in v nor that in w preserves the relevant lattice, as $2 \text{div}(v)/|v^2| = 2/3$ and $2 \text{div}(w)/|w^2| = 1/2$. Thus the pullbacks to the period domains have multiplicity one.

The Borchers products have these same reduced divisors and weights. The quotients are consequently nowhere-vanishing holomorphic functions of weight zero, possibly with a finite character. Passing to a finite-index subgroup kills that character. The Baily–Borel boundary

has codimension at least two in both cases, so normality extends these functions across the boundary. Compactness makes them constant. In particular the characters agree, and the proportionalities hold for the original groups. $\square$

Finally, let J_7 and J_6 denote the modular Jacobians of the ordered generator systems in Equations (95) and (96). Their weights are

$$\text{wt}(J_7) = (4 + 6 + 10 + 12 + 14 + 16 + 18 + 22 + 24 + 30) + 9 = 165,$$

and

$$\text{wt}(J_6) = (4 + 6 + 7 + 10 + 12 + 15 + 16 + 18 + 24) + 8 = 120.$$

The modular-Jacobian criterion and the strongly reflective products of Gritsenko–Nikulin therefore give

$$(118) \quad J_7 \doteq \Phi_{165, E_7}, \quad J_6 \doteq \Phi_{120, E_6};$$

see [28, 69]. The four numbers 44, 45, 165, 120 that arise from the local discriminants and coefficient Jacobians are exactly the exceptional Borcherds weights.

Remark 5.4. Sakai’s E_7 and E_6 Seiberg–Witten curves realize the Wirthmüller generators as Weyl-invariant weak Jacobi forms [55]. Wang–Williams obtain the orthogonal generators in (95) and (96) by additive lifting of the corresponding holomorphic Jacobi forms. After matching Fourier–Jacobi normalizations, Equations (104) and (108) provide a K3-geometric orthogonal lift of the same exceptional deformation data. The arithmetic progressions of coefficient weights follow directly from homogeneity: every monomial in the E_7 equation has weight 42, and the two odd E_6 coefficients enter through the quadratic invariants in (107).

6. WITT RESTRICTION AND THE INOSE FAMILY

Put $L_E = U^{\oplus 2} \oplus E_8(-1)$. At the standard one-dimensional cusp, write a Fourier–Jacobi expansion in tube-domain coordinates as

$$F(\tau_1, z, \tau_2) = \sum_{m \geq 0} \phi_m(\tau_1, z) q_2^m, \quad q_2 = e^{2\pi i \tau_2}.$$

The Witt restriction is evaluation on the subdomain $z = 0$:

$$(119) \quad W(F)(\tau_1, \tau_2) = F(\tau_1, 0, \tau_2) = \sum_{m \geq 0} \phi_m(\tau_1, 0) q_2^m.$$

The orthogonal group interchanges the two copies of U , so the image is symmetric in τ_1, τ_2 . Let E_4, E_6 be the normalized elliptic Eisenstein series and put

$$(120) \quad e_4 = E_4(\tau_1)E_4(\tau_2), \quad e_6 = E_6(\tau_1)E_6(\tau_2), \quad d = \Delta(\tau_1)\Delta(\tau_2),$$

where $\Delta = (E_4^3 - E_6^2)/1728$. These have weights 4, 6, 12, respectively. The geometric interpretation of this diagonal restriction uses the classical Shioda–Inose construction and its lattice-polarized formulation [63, 32, 46, 7].

Proposition 6.1. *The graded ring of symmetric diagonal bi-modular forms is*

$$(121) \quad \bigoplus_{k \geq 0} (M_k(\text{SL}_2(\mathbb{Z})) \otimes M_k(\text{SL}_2(\mathbb{Z})))^{\mathfrak{S}_2} = \mathbb{C}[e_4, e_6, d].$$

Consequently, W maps the Hashimoto–Ueda ring to the polynomial ring on the right.

Proof. Write $x_i = E_4(\tau_i)$, $y_i = E_6(\tau_i)$, and $D = 1728^2 d$. The two mixed monomials $A = x_1^3 y_2^2$ and $B = x_2^3 y_1^2$ satisfy

$$A + B = e_4^3 + e_6^2 - D, \quad AB = e_4^3 e_6^2.$$

Every symmetric pair of monomials of equal weight is a monomial in e_4, e_6 times $A^r + B^r$ for some $r \geq 0$. The power sums $A^r + B^r$ are generated recursively by $A + B$ and AB . This proves generation. The diagonal subring of $\mathbb{C}[x_1, y_1, x_2, y_2]$ has Krull dimension three, and taking the finite $\mathfrak{S}_2$ -invariants preserves dimension. Since it is a domain generated by the three elements e_4, e_6, d , they are algebraically independent. This is also the symmetric split-Hilbert-modular description of the $U^{\oplus 2}$ -subdomain. $\square$

Dieckmann, Krieg, and Woitalla proved that the normalized orthogonal Eisenstein series $\mathcal{E}_k$, for

$$k = 4, 10, 12, 16, 18, 22, 24, 28, 30, 36, 42,$$

form another system of polynomial generators for $M_*(\mathcal{O}^+(L_E))$ [12, Theorem 4.3]. Thus the Hashimoto–Ueda coefficients and these Eisenstein series are two coordinate systems on the same affine cone. Substitution in that change of generators and then in (119) gives the following result.

Theorem 6.2 (Witt–Inose restriction). *Set*

$$(122) \quad a = -3e_4, \quad b = \frac{2}{7}e_6, \quad D = 1728^2 d.$$

The Witt restrictions of the Hashimoto–Ueda generators are

$$(123) \quad \begin{aligned} W(C_4) &= a, & W(C_{10}) &= -4ab, \\ W(C_{12}) &= -21b^2 + D, & W(C_{16}) &= 6ab^2, \\ W(C_{18}) &= 70b^3 - 5bD, & W(C_{22}) &= -4ab^3, \\ W(C_{24}) &= -105b^4 + 10b^2D, & W(C_{28}) &= ab^4, \\ W(C_{30}) &= 84b^5 - 10b^3D, & W(C_{36}) &= -35b^6 + 5b^4D, \\ W(C_{42}) &= 6b^7 - b^5D. \end{aligned}$$

The Witt restriction of the universal family (78), after change of variables, becomes

$$(124) \quad y^2 = x^3 - 3e_4 T^4 x + T^5 (T^2 + 2e_6 T + 1728^2 d).$$

For $d \neq 0$, Equation (124) is the modular Inose family associated with the unordered pair (E_{τ_1}, E_{τ_2}) . Its minimal resolution is generically polarized by $U \oplus E_8(-1)^{\oplus 2}$, and its standard elliptic fibration has two fibers of type II^ .*

The divisor $d = 0$ in the Baily–Borel compactification of the Witt surface maps exactly onto the Hashimoto–Ueda non- $K3$ locus. In particular, on $d = 0$ the coefficient point is

$$(125) \quad [\mathbf{C}] = [a : -4ab : -21b^2 : 6ab^2 : 70b^3 : -4ab^3 : -105b^4 : ab^4 : 84b^5 : -35b^6 : 6b^7].$$

Proof. The identities (123) follow by restricting the Eisenstein polynomials and using $E_4^3 - E_6^2 = 1728\Delta$. Equivalently, the resulting coefficient substitution in (78) is

$$(126) \quad y^2 = x^3 + a(t - b)^4 x + (t - b)^5 ((t - b)^2 + 7b(t - b) + D).$$

After the translation $T = t - b$, this becomes (124). Expanding (126) gives exactly those eleven coefficients; conversely, collecting the displayed coefficients yields the factorization.

For $d \neq 0$, the orders at $T = 0$ are $(\geq 4, 5, 10)$ for $(f, g, 4f^3 + 27g^2)$, and the same orders occur at $T = \infty$. Hence the two distinguished fibers are of type II^* . Moreover,

$$(127) \quad j(\tau_1)j(\tau_2) = \frac{e_4^3}{d}, \quad (j(\tau_1) - 1728)(j(\tau_2) - 1728) = \frac{e_6^2}{d}.$$

Thus the two classical j -invariants, up to permutation, are recovered from (e_4, e_6, d) . The standard Shioda–Inose construction identifies the resulting $U \oplus E_8(-1)^{\oplus 2}$ -polarized K3 surface with the Inose surface of $E_{\tau_1} \times E_{\tau_2}$ [63, 32, 46, 7].

Finally, $d = 0$ is the image of $\Delta(\tau_1)\Delta(\tau_2) = 0$. Before quotienting by $\mathfrak{S}_2$ this has two components, which are interchanged by permutation; after quotienting it is the single one-dimensional Baily–Borel boundary component $\text{Proj } \mathbb{C}[e_4, e_6]$. Setting $D = 0$ gives (125) and in the Weierstrass model

$$(128) \quad f(t) = a(t - b)^4, \quad g(t) = (t - b)^6(t + 6b).$$

At $t = b$ one has $\text{ord}(f, g) \geq (4, 6)$, so the Weierstrass model is non-minimal and is not a K3 surface. Hashimoto and Ueda prove that (125) is the entire non-K3 locus [30, Corollary 3.2]; hence the image is exact. $\square$

Remark 6.3. The Witt map (119) is restriction to a two-dimensional totally geodesic modular surface, not itself a boundary map. Its divisor $d = 0$ is the one-dimensional Baily–Borel boundary, and the intersection of its two preimages is the zero-dimensional cusp; compare the general description of orthogonal Baily–Borel boundary components in [59, 14].

7. FIVE-DIMENSIONAL GAUSSIAN BALL SUBFAMILIES

Let L be one of the rank-ten lattices M_i from $U \oplus E_8(-2)$ through $U \oplus E_8(-1)$, and put $T_L = L^\perp \subset \Lambda_{\text{K3}}$. On the locus considered here, T_L carries an isometry ρ_T with $\rho_T^2 = -1$. The i -eigenspace has dimension six and Hermitian signature $(1, 5)$; projectivizing its positive cone gives a complex ball $\mathbb{B}^5$. We restrict to the geometric locus on which ρ_T extends to an order-four Hodge isometry of the K3 lattice that preserves an ample chamber. By the global Torelli theorem, the extension is then induced by a purely non-symplectic automorphism ρ of order four. This cuts the ten-dimensional type-IV moduli space to a five-dimensional unitary modular subvariety, without increasing the Picard rank of a very general member [37, 9, 2].

We first restrict the five Jacobian presentations in (7) to the complex ball, and derive the corresponding equations and automorphisms. The standard residue form shows that the displayed maps act by $\pm i$ on $H^{2,0}$.

- (i) For $U \oplus E_8(-2)$, quadratic base change of a rational elliptic surface and the Gaussian restriction give

$$(129) \quad \begin{aligned} y^2 &= x^3 + (f_0 + f_2\xi^4 + f_4\xi^8)x + g_0 + g_2\xi^4 + g_4\xi^8 + g_6\xi^{12}, \\ \rho(\xi, x, y) &= (i\xi, x, y). \end{aligned}$$

- (ii) For $U \oplus N$, one has

$$(130) \quad \begin{aligned} Y^2 &= X^3 + b(t)X, \quad \deg b = 8, \\ \rho(t, X, Y) &= (t, -X, iY). \end{aligned}$$

If b has eight simple roots, the singular fibers are eight fibers of type III; their unordered positions give five moduli.

(iii) For $U \oplus D_4(-1)^{\oplus 2}$, one has

$$(131) \quad \begin{aligned} y^2 &= x^3 + u^2(a_0 + a_2u^2 + a_4u^4)x + u^3(b_0 + b_2u^2 + b_4u^4 + b_6u^6), \\ \rho(u, x, y) &= (-u, -x, iy). \end{aligned}$$

(iv) For $U \oplus D_8(-1)$, one has

$$(132) \quad \begin{aligned} y^2 &= x^3 + (s^3 + sF_4)x^2 + (s^2G_8 + G_{12})x + H_8s^5 + H_{12}s^3 + H_{16}s, \\ \rho(s, x, y) &= (-s, -x, iy). \end{aligned}$$

(v) For $U \oplus E_8(-1)$, the Hashimoto–Ueda model restricts to

$$(133) \quad \begin{aligned} y^2 &= x^3 + (C_4t^4 + C_{16}t^2 + C_{28})x + t^7 + C_{12}t^5 + C_{24}t^3 + C_{36}t, \\ \rho(t, x, y) &= (-t, -x, iy). \end{aligned}$$

In (i), ρ^2 is the deck involution of the quadratic base change; in (ii)–(v), it is the fiberwise elliptic involution.

7.1. Kondō’s eight-point space and the restricted D_8 coefficients. Let $\text{Conf}_8(\mathbb{P}^1) \subset (\mathbb{P}^1)^8$ be the locus of ordered eight-tuples of pairwise distinct points, and put

$$\mathcal{M}_K^{\text{ord}} := \text{Conf}_8(\mathbb{P}^1) / \text{PGL}_2, \quad \overline{\mathcal{M}}_K^{\text{ord}} := (\mathbb{P}^1)^8 //_{\mathcal{O}(1, \dots, 1)} \text{SL}_2.$$

Thus $\mathcal{M}_K^{\text{ord}} = M_{0,8}$, while $\overline{\mathcal{M}}_K^{\text{ord}}$ is the GIT compactification by semistable ordered configurations. Kondō identifies the latter with the Satake–Baily–Borel compactification of $\mathbb{B}^5/\Gamma(1-i)$, where Γ is his ambient arithmetic unitary group and $\Gamma(1-i)$ is its principal congruence subgroup of level $1-i$. The identification is S_8 -equivariant, and $\Gamma/\Gamma(1-i) \cong S_8$ [37, Theorems 3.7 and 4.11]. We write

$$\mathcal{M}_K := \mathcal{M}_K^{\text{ord}} / S_8$$

for the corresponding unlabelled moduli space. This convention distinguishes Kondō’s ordered ball quotient from the moduli space of the underlying unlabelled K3 surfaces; compare [41, 40].

Write a point of $\mathbb{P}^1$ as a nonzero column vector $v_i \in \mathbb{C}^2$, and put $[ij] := \det(v_i, v_j)$. A tableau in this setting is a perfect matching of $\{1, \dots, 8\}$, represented by four rows (τ_{r1}, τ_{r2}) , and its bracket monomial is

$$(134) \quad \mu_\tau := \prod_{r=1}^4 [\tau_{r1} \tau_{r2}].$$

Reversing the orientation of one row changes the sign of μ_τ ; the conventional orientations of standard tableaux fix these signs. There are $8!/(2^4 4!) = 105$ matching classes. Straightening by the Plücker relations gives fourteen standard tableaux $\tau_1, \dots, \tau_{14}$, which form a basis of the degree-one invariants and define the classical map $\overline{\mathcal{M}}_K^{\text{ord}} \rightarrow \mathbb{P}^{13}$; see [13, 31]. If F_V is Kondō’s weight-four orthogonal modular form attached to the corresponding maximal totally singular subspace V , then $F_V|_{\mathbb{B}^5}$ has an even divisor. Since $\mathbb{B}^5$ is simply connected, Kondō

chooses a square root G_V of ball weight two. For the subspaces $V_1, \dots, V_{14}$ corresponding to the chosen standard tableaux, his comparison theorem states

$$(135) \quad [G_{V_1} : \dots : G_{V_{14}}] = [\mu_{\tau_1} : \dots : \mu_{\tau_{14}}].$$

This is an equality of projective maps, not a canonical affine equality between individual modular forms and bracket polynomials [37, Sections 6–7, especially Theorem 7.5].

The finite data required by the neighbor construction consist of an ordered decomposition $S_0 \sqcup S_\infty = \{1, \dots, 8\}$, with $|S_0| = |S_\infty| = 4$, and a perfect matching of each four-element set. Let $\mathfrak{d}_0$ denote the reference datum

$$S_0 = \{1, 2, 3, 4\}, \quad S_\infty = \{5, 6, 7, 8\}, \quad \pi_0 = (12)(34), \quad \pi_\infty = (56)(78),$$

and let $H_{\mathfrak{d}_0} \subset S_8$ be its stabilizer. We define

$$(136) \quad \widetilde{\mathcal{M}}_K := \mathcal{M}_K^{\text{ord}} / H_{\mathfrak{d}_0} \longrightarrow \mathcal{M}_K = \mathcal{M}_K^{\text{ord}} / S_8.$$

It is finite of generic degree $\binom{8}{4} \cdot 3^2 = 630$, and it is étale over the locus of configurations with trivial projective automorphism group. It remembers precisely the ordered $4 + 4$ decomposition and the two pairings. Let $\varepsilon_{\tau_0} : H_{\mathfrak{d}_0} \rightarrow \{\pm 1\}$ be the character by which the stabilizer acts on μ_{τ_0} , and put $H_{\mathfrak{d}_0}^+ := \ker(\varepsilon_{\tau_0})$. The additional character cover is

$$\widetilde{\mathcal{M}}_K^{H_4} := \mathcal{M}_K^{\text{ord}} / H_{\mathfrak{d}_0}^+ \longrightarrow \widetilde{\mathcal{M}}_K.$$

On this twofold cover the sign of H_4 is defined. An affine coordinate on $\mathbb{P}^1$, monic equations for the two quartics, and a Weierstrass scale are coordinate gauges rather than additional finite marking data.

On the affine chart on which all branch points are finite, write

$$(137) \quad b_1(t) = \prod_{j=1}^4 (t - \lambda_j), \quad b_2(t) = \prod_{j=5}^8 (t - \lambda_j).$$

We henceforth choose the affine lifts $v_i = (1, \lambda_i)^t$, so that $[ij] = \lambda_j - \lambda_i$. The restricted double-cover model is

$$(138) \quad Z^2 = b_2(t)\xi^4 + b_1(t).$$

Put $u = \xi^2$ and $v = \xi Z$. Then the quartic admits the following degree-two map to an elliptic curve:

$$(139) \quad v^2 = u(b_2(t)u^2 + b_1(t)), \quad (X, Y) = (b_2(t)u, b_2(t)v), \quad Y^2 = X^3 + b_1(t)b_2(t)X.$$

The first equation is the affine double-cover model used by Kondō, and the last equation is its birational Weierstrass form.¹ The map $(\xi, Z) \mapsto (u, v)$ is the fiberwise degree-two covering over the t -line. In the coefficient construction below we change the ruling: u is the base coordinate and t is the quartic fiber coordinate. The Weierstrass form has constant j -invariant 1728 and discriminant proportional to $(b_1 b_2)^3$ [37, Section 2].

¹The relative Jacobian of (138) over the t -line is $y^2 = x^3 - 4b_1(t)b_2(t)x$; its degree-two isogenous curve is the last equation in (139). Over $\mathbb{C}(t)$ these two $j = 1728$ curves are isomorphic after a constant rescaling, but the displayed degree-two map is not that isomorphism.

Proposition 7.1. *Fix the affine coordinate and lifts, the monic quartics in (137), and the marking $\mathfrak{d}_0$. The two-neighbor and relative-Jacobian construction defines a polynomial map*

$$(\lambda_1, \dots, \lambda_8) \longmapsto (F_4, G_8, H_8, G_{12}, H_{12}, H_{16}).$$

With $\deg(\lambda_i) = 1$, their ordinary total degrees are

$$(140) \quad \begin{array}{c|cccccc} \text{coefficient} & F_4 & G_8 & H_8 & G_{12} & H_{12} & H_{16} \\ \hline \deg_\lambda & 4 & 8 & 8 & 12 & 12 & 16 \end{array}$$

In particular, for $\tau_0 = (12)(34)(56)(78)$ one has

$$(141) \quad H_8 = -\frac{1}{4}[12]^2[34]^2[56]^2[78]^2 = -\frac{1}{4}\mu_{\tau_0}^2,$$

and one can choose $H_4 = \mu_{\tau_0}/2$ on $\widetilde{\mathcal{M}}_K^{H_4}$.

Proof. Put $Q(t, u) := b_1(t) + u^2b_2(t)$, and write $Q(t, u) = A_0(u)t^4 + A_1(u)t^3 + A_2(u)t^2 + A_3(u)t + A_4(u)$. We use the classical binary-quartic invariants to compute its relative Jacobian fibration. Since every $A_i(u)$ is affine-linear in u^2 , there are polynomials I_{2j} and J_{2j} in the λ_i such that

$$\mathcal{I}(Q) = I_0 + I_2u^2 + I_4u^4, \quad \mathcal{J}(Q) = J_0 + J_2u^2 + J_4u^4 + J_6u^6.$$

Set $a_{2j} := -I_{2j}/3$ and $b_{2j} := -J_{2j}/27$. The relative Jacobian of $v^2 = uQ(t, u)$, regarded as a genus-one fibration over the u -line, is

$$(142) \quad y^2 = x^3 + u^2(a_0 + a_2u^2 + a_4u^4)x + u^3(b_0 + b_2u^2 + b_4u^4 + b_6u^6).$$

This is the restricted $U \oplus D_4(-1)^{\oplus 2}$ model.

For a four-element set S , let $e_2(S) := \sum_{i < j, i, j \in S} \lambda_i \lambda_j$, and let $\text{Pair}(S)$ be its set of three perfect matchings. If $\pi = \{\{i, j\}, \{k, l\}\} \in \text{Pair}(S)$, define

$$\gamma_\pi(S) := \frac{1}{3}e_2(S) - (\lambda_i \lambda_j + \lambda_k \lambda_l), \quad \alpha_\pi := \gamma_\pi(\{1, 2, 3, 4\}), \quad \beta_\pi := \gamma_\pi(\{5, 6, 7, 8\}).$$

A computation gives

$$z^3 + a_0z + b_0 = \prod_{\pi \in \text{Pair}(\{1, 2, 3, 4\})} (z - \alpha_\pi), \quad z^3 + a_4z + b_6 = \prod_{\pi \in \text{Pair}(\{5, 6, 7, 8\})} (z - \beta_\pi).$$

Thus, the three perfect matchings of the first four points give the three possible roots α , and the three perfect matchings of the last four points give the three possible roots β . For a generic configuration there are therefore $3 \cdot 3 = 9$ possible pairs (α, β) . The reference marking gives

$$(143) \quad \alpha = \frac{1}{3}e_2(\{1, 2, 3, 4\}) - (\lambda_1 \lambda_2 + \lambda_3 \lambda_4), \quad \beta = \frac{1}{3}e_2(\{5, 6, 7, 8\}) - (\lambda_5 \lambda_6 + \lambda_7 \lambda_8).$$

Substitute

$$x = u(\alpha + su + \beta u^2), \quad y = u^2w$$

into (142). Because $\alpha^3 + a_0\alpha + b_0 = 0$ and $\beta^3 + a_4\beta + b_6 = 0$, division by the common power of u leaves the quartic

$$(144) \quad w^2 = ps + (qs^2 + r)u + (s^3 + ms)u^2 + (ns^2 + o)u^3 + hsu^4,$$

where we have introduced

$$(145) \quad \begin{aligned} p &= 3\alpha^2 + a_0, & q &= 3\alpha, & r &= \beta p + a_2\alpha + b_2, \\ m &= 6\alpha\beta + a_2, & n &= 3\beta, & o &= 3\alpha\beta^2 + a_2\beta + a_4\alpha + b_4, \\ h &= 3\beta^2 + a_4. \end{aligned}$$

For the reference pairings they also satisfy the identities

$$\begin{aligned} p &= [13][14][23][24], & q &= -([13][24] + [14][23]), \\ h &= [57][58][67][68], & n &= -([57][68] + [58][67]). \end{aligned}$$

Applying the same binary-quartic invariants to the right-hand side of (144) gives

$$\tilde{\mathcal{I}} = s^6 + 2F_4s^4 + \tilde{I}_2s^2 - 3or, \quad \tilde{\mathcal{J}} = -2s^9 - 6F_4s^7 + \tilde{J}_5s^5 + \tilde{J}_3s^3 + \tilde{J}_1s,$$

where we have set

$$\begin{aligned} \tilde{I}_2 &= 12hp - 3(nr + oq) + m^2, \\ \tilde{J}_5 &= 72hp + 9(nr + nmq + oq) - 27hq^2 - 27pn^2 - 6m^2, \\ \tilde{J}_3 &= 72hpm + 9(nmr + or + omq) - 54hqr - 54pno - 2m^3, \\ \tilde{J}_1 &= 9omr - 27hr^2 - 27po^2. \end{aligned}$$

Comparing the Weierstrass equation $y^2 = x^3 - \tilde{\mathcal{I}}x/3 - \tilde{\mathcal{J}}/27$ with (132) yields

$$(146) \quad \begin{aligned} F_4 &= a_2 - \frac{15}{2}\alpha\beta, & G_8 &= \frac{F_4^2 - \tilde{I}_2}{3}, & G_{12} &= or, \\ H_8 &= -\frac{\tilde{J}_5}{27} - \frac{F_4^2 + \tilde{I}_2}{9}, & H_{12} &= -\frac{\tilde{J}_3}{27} - \frac{2F_4^3}{27} + \frac{G_{12} + F_4G_8}{3}, & H_{16} &= -\frac{\tilde{J}_1}{27} + \frac{F_4G_{12}}{3}. \end{aligned}$$

Every coefficient of b_1 and b_2 is an elementary symmetric polynomial in the appropriate four branch points. Hence $I_{2j}, J_{2j}, \alpha, \beta$, the seven coefficients in (145), and finally the six expressions in (146) are polynomials in the λ_i . Assigning degree one to each λ_i , one has

$$\deg(a_0, a_2, a_4) = 4, \quad \deg(b_0, b_2, b_4, b_6) = 6, \quad \deg(\alpha) = \deg(\beta) = 2,$$

from which (140) follows. Finally, substituting the closed formulas and using the two end-point cubic relations gives

$$H_8 = -\frac{1}{4}(3\alpha^2 + 4a_0)(3\beta^2 + 4a_4).$$

The endpoint bracket identities

$$3\alpha^2 + 4a_0 = -[12]^2[34]^2, \quad 3\beta^2 + 4a_4 = -[56]^2[78]^2$$

then give (141). □

APPENDIX A. WEIERSTRASS EQUATIONS ASSOCIATED WITH ROOT SYSTEMS

Table 1 collects Weierstrass models for families whose very general member has transcendental lattice $T_R = U^{\oplus 2} \oplus L_R(-1)$, where U is the hyperbolic plane (also denoted by H) and L_R is the root lattice in its standard even normalization. For simply laced R one has $L_R = R$; in the other cases the conventions are

$$L_{B_r} = A_1^{\oplus r}, \quad L_{C_r} = D_r, \quad L_{G_2} = A_2, \quad L_{F_4} = D_4,$$

with $D_3 \cong A_3$. The modular group is

$$\Gamma_R = \langle \tilde{O}^+(T_R), W(R) \rangle,$$

where $\tilde{O}^+(T_R)$ is the discriminant kernel and $W(R)$ acts trivially on $U^{\oplus 2}$; see [69, Theorems 1.2 and 5.9], and [30] for E_8 . The column $\rho = 18 - \text{rank}(R)$ gives the Picard rank of the very general member.

The subscripts indicate modular weights, and distinct letters with the same subscript denote distinct generators. The generators of $M_*(\Gamma_R)$ occur in the combinations displayed in the table, with row-dependent normalizations; the D_4 model reflects triality. In the A_r and E_6 models, the relevant square terms have been moved to the left-hand side so that the odd-weight generators occur linearly. The base coordinate is t throughout. In the E_6 row, $c_7 = 2Q_7$, $c_{15} = 2Q_{15}$, and $c_k = Q_k$ for the even weights; this agrees with (109) after renaming coordinates. The E_7 and E_8 rows use $c_k = P_k$ and $c_k = C_k$, respectively.

Table 1: Weierstrass models associated with irreducible root systems.

R	ρ	Weierstrass equation
A_1	17	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + (g_{10}t + g_{12})x$
A_2	16	$y^2 + g_9y = x^3 + (t^3 + f_4t + f_6)x^2 + (g_{10}t + g_{12})x$
B_2	16	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + (g_8t^2 + g_{10}t + g_{12})x$
G_2	16	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + (g_{10}t + g_{12})x + h_{18}$
A_3	15	$y^2 + g_9y = x^3 + (t^3 + f_4t + f_6)x^2 + (g_8t^2 + g_{10}t + g_{12})x$
B_3	15	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + (g_6t^3 + g_8t^2 + g_{10}t + g_{12})x$
C_3	15	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + (g_8t^2 + g_{10}t + g_{12})x + h_{18}$
A_4	14	$y^2 + (g_7t + g_9)y = x^3 + (t^3 + f_4t + f_6)x^2 + (g_8t^2 + g_{10}t + g_{12})x$
B_4	14	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + (g_4t^4 + g_6t^3 + g_8t^2 + g_{10}t + g_{12})x$
C_4	14	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + (g_8t^2 + g_{10}t + g_{12})x + h_{16}t + h_{18}$
D_4	14	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + ((a_8 - b_8)t^2 + g_{10}t + g_{12})x - a_8b_8t + h_{18}$

Continued on next page

Table 1 (continued)

R	ρ	Weierstrass equation
F_4	14	$y^2 = x^3 + (f_4t^4 + g_{10}t^3 + \chi_{16}t^2)x$ $+ (t^7 + f_6t^6 + g_{12}t^5 + h_{18}t^4 + \chi_{24}t^3)$
A_5	13	$y^2 + (g_7t + g_9)y = x^3 + (t^3 + f_4t + f_6)x^2$ $+ (g_6t^3 + g_8t^2 + g_{10}t + g_{12})x$
C_5	13	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + (g_8t^2 + g_{10}t + g_{12})x$ $+ h_{14}t^2 + h_{16}t + h_{18}$
D_5	13	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + (g_8t^2 + g_{10}t + g_{12})x$ $+ g_7^2t^2 + h_{16}t + h_{18}$
A_6	12	$y^2 + (g_5t^2 + g_7t + g_9)y$ $= x^3 + (t^3 + f_4t + f_6)x^2 + (g_6t^3 + g_8t^2 + g_{10}t + g_{12})x$
C_6	12	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + (g_8t^2 + g_{10}t + g_{12})x$ $+ h_{12}t^3 + h_{14}t^2 + h_{16}t + h_{18}$
D_6	12	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + (g_8t^2 + g_{10}t + g_{12})x$ $- g_6^2t^3 + h_{14}t^2 + h_{16}t + h_{18}$
E_6	12	$y^2 + c_7xy + (c_{15}t)y$ $= x^3 + (c_4t^4 + c_{10}t^3 + c_{16}t^2)x$ $+ t^7 + c_6t^6 + c_{12}t^5 + c_{18}t^4 + c_{24}t^3$
A_7	11	$y^2 + (g_5t^2 + g_7t + g_9)y$ $= x^3 + (t^3 + f_4t + f_6)x^2 + (g_4t^4 + g_6t^3 + g_8t^2 + g_{10}t + g_{12})x$
C_7	11	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + (g_8t^2 + g_{10}t + g_{12})x$ $+ h_{10}t^4 + h_{12}t^3 + h_{14}t^2 + h_{16}t + h_{18}$
D_7	11	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + (g_8t^2 + g_{10}t + g_{12})x$ $+ g_5^2t^4 + h_{12}t^3 + h_{14}t^2 + h_{16}t + h_{18}$
E_7	11	$y^2 = x^3 + c_{14}x^2 + (c_4t^4 + c_{10}t^3 + c_{16}t^2 + c_{22}t)x$ $+ t^7 + c_6t^6 + c_{12}t^5 + c_{18}t^4 + c_{24}t^3 + c_{30}t^2$
C_8	10	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + (g_8t^2 + g_{10}t + g_{12})x$ $+ h_8t^5 + h_{10}t^4 + h_{12}t^3 + h_{14}t^2 + h_{16}t + h_{18}$
D_8	10	$y^2 = x^3 + (t^3 + f_4t + f_6)x^2 + (g_8t^2 + g_{10}t + g_{12})x$ $- g_4^2t^5 + h_{10}t^4 + h_{12}t^3 + h_{14}t^2 + h_{16}t + h_{18}$
E_8	10	$y^2 = x^3 + (c_4t^4 + c_{10}t^3 + c_{16}t^2 + c_{22}t + c_{28})x$ $+ t^7 + c_{12}t^5 + c_{18}t^4 + c_{24}t^3 + c_{30}t^2 + c_{36}t + c_{42}$

APPENDIX B. A TWO-NEIGHBOR CONSTRUCTION FOR $U \oplus D_4(-1)^{\oplus 2} \cong U(2) \oplus D_8(-1)$

We describe the two-neighbor construction which transforms the Jacobian elliptic fibration realizing the lattice polarization $U \oplus D_4(-1)^{\oplus 2}$ into a genus-one fibration realizing the polarization $U(2) \oplus D_8(-1)$. Let X be a K3 surface given by the Weierstrass equation

$$(147) \quad \begin{aligned} y^2 = & x^3 + u^2(a_0 + a_1u + a_2u^2 + a_3u^3 + a_4u^4)x \\ & + u^3(b_0 + b_1u + b_2u^2 + b_3u^3 + b_4u^4 + b_5u^5 + b_6u^6). \end{aligned}$$

For a general choice of the coefficients, the Jacobian elliptic fibration $\pi : X \rightarrow \mathbb{P}_{(u)}^1$ has singular fibers $2I_0^* + 12I_1$ and trivial Mordell–Weil group. We construct a second fibration on X by choosing suitable components of the two I_0^* fibers. At $u = 0$, the leading cubic associated with the I_0^* fiber is

$$(148) \quad z^3 + a_0z + b_0,$$

and we choose a root α of this cubic. Similarly, at $u = \infty$, the leading cubic is

$$(149) \quad z^3 + a_4z + b_6,$$

and we choose a root β . Geometrically, the choices of α and β determine distinguished simple components in the two fibers. We introduce a new parameter s by considering the pencil

$$(150) \quad x = u(\alpha + su + \beta u^2) \quad \Leftarrow \quad s = \frac{x - \alpha u - \beta u^3}{u^2}.$$

The choice of the linear terms αu and βu^3 forces the pencil to have the prescribed incidence with the chosen components of the fibers over $u = 0$ and $u = \infty$. This is what reduces the sextic model to a quartic. Substituting

$$x = u(\alpha + su + \beta u^2), \quad y = u^2w$$

into (147), we obtain

$$u^4w^2 = u^3 \left[(\alpha + su + \beta u^2)^3 + (a_0 + \cdots + a_4u^4)(\alpha + su + \beta u^2) + (b_0 + \cdots + b_6u^6) \right],$$

whence

$$(151) \quad w^2 = \frac{(\alpha + su + \beta u^2)^3 + (a_0 + \cdots + a_4u^4)(\alpha + su + \beta u^2) + (b_0 + \cdots + b_6u^6)}{u}.$$

Because α and β are roots of Equations (148) and (149), respectively, the numerator in (151) has vanishing constant term and vanishing coefficient of u^6 . Therefore, the right-hand side is a polynomial of degree four in u . Equation (151) defines a genus-one fibration $\varphi : X \rightarrow \mathbb{P}_{(s)}^1$ whose generic fiber is a double cover of the u -line branched over a quartic. The new fibration is given by

$$(152) \quad w^2 = A_4(s) + A_3(s)u + A_2(s)u^2 + A_1(s)u^3 + A_0(s)u^4,$$

with

$$\begin{aligned}
(153) \quad & A_4(s) = (3\alpha^2 + a_0)s + a_1\alpha + b_1, \\
& A_3(s) = 3\alpha s^2 + a_1s + 3\alpha^2\beta + a_0\beta + a_2\alpha + b_2, \\
& A_2(s) = s^3 + (6\alpha\beta + a_2)s + a_1\beta + a_3\alpha + b_3, \\
& A_1(s) = 3\beta s^2 + a_3s + 3\alpha\beta^2 + a_2\beta + a_4\alpha + b_4, \\
& A_0(s) = (3\beta^2 + a_4)s + a_3\beta + b_5.
\end{aligned}$$

The old elliptic fiber class intersects the fibers of φ in degree two and therefore supplies a bisection. To verify the remaining assertions, apply the binary-quartic invariants $\mathcal{I}(s)$ and $\mathcal{J}(s)$ to (152). The relative-Jacobian invariants show that the fibration has a fiber of type I_4^* at infinity, while the fourteen simple finite roots of the discriminant give fourteen fibers of type I_1 [44].

For the very general member, the original Jacobian fibration has two fibers of type I_0^* , twelve fibers of type I_1 , and trivial Mordell–Weil group. The Shioda–Tate formula then gives [62]

$$\mathrm{NS}(X) \cong U \oplus D_4(-1)^{\oplus 2} \cong U(2) \oplus D_8(-1).$$

The bisection shows that the divisibility of the new primitive fiber class is either one or two. If it were one, then φ would have a section, and its fiber, zero section, and I_4^* -components would generate the full-rank trivial lattice $U \oplus D_8(-1)$, of discriminant 4, inside $\mathrm{NS}(X)$, of discriminant 16. This is impossible because a finite-index inclusion of lattices satisfies

$$|\mathrm{disc}(U \oplus D_8(-1))| = [\mathrm{NS}(X) : U \oplus D_8(-1)]^2 |\mathrm{disc} \mathrm{NS}(X)|.$$

Thus the new fiber class has divisibility two. Hence φ has no section for the very general member, and the old fiber class is its natural bisection.

APPENDIX C. THE HASHIMOTO–UEDA FAMILY

C.1. Orthogonal modular forms and the modular Jacobian. Let L be an even lattice of signature $(2, n)$, let $\Gamma \subset \mathrm{O}^+(L)$ be an arithmetic subgroup, and let $\mathcal{D}_L$ be a connected component of

$$\{[Z] \in \mathbb{P}(L \otimes \mathbb{C}) : (Z, Z) = 0, (Z, \bar{Z}) > 0\},$$

and let

$$\mathcal{A}_L = \{Z \in L \otimes \mathbb{C} : [Z] \in \mathcal{D}_L\}$$

be its affine cone. Fix a primitive isotropic vector $c \in L$ and the corresponding affine Siegel-domain slice

$$\mathcal{S}_c = \{Z \in \mathcal{A}_L : (Z, c) = 1\},$$

and fix tube-domain coordinates $z_1, \dots, z_n$ arising from a chosen linear basis of $c^\perp/c$ at a cusp on $\mathcal{S}_c$. Suppose that $f_0, \dots, f_n$ are modular forms of weights $k_0, \dots, k_n$ and characters $\chi_0, \dots, \chi_n$. Their modular Jacobian is defined to be

$$(154) \quad J(f_0, \dots, f_n) = \det \begin{pmatrix} k_0 f_0 & k_1 f_1 & \cdots & k_n f_n \\ \partial_{z_1} f_0 & \partial_{z_1} f_1 & \cdots & \partial_{z_1} f_n \\ \vdots & \vdots & & \vdots \\ \partial_{z_n} f_0 & \partial_{z_n} f_1 & \cdots & \partial_{z_n} f_n \end{pmatrix}.$$

With the convention $f(\lambda Z) = \lambda^{-k} f(Z)$, the first row is the negative of the radial Euler derivative on $\mathcal{A}_L$. The sign is chosen to match the standard modular-Jacobian convention. Replacing the chosen linear tube-domain basis changes the determinant only by a nonzero constant.

One has the following:

Proposition C.1. *The determinant (154) is a modular form of weight*

$$k_J = k_0 + \cdots + k_n + n$$

and character

$$\chi_J = \chi_0 \cdots \chi_n \det.$$

If all χ_j are trivial and a reflection $\sigma_r \in \Gamma$ fixes the hyperplane $r^\perp$, then $J(f_0, \dots, f_n)$ vanishes on $r^\perp$. If, moreover, the scalar-valued ring is freely generated by the f_j and the reflection criterion applies, then the modular Jacobian has a simple zero along each reflective hyperplane and generates the module of forms with determinant character.

Proof. Let $j(\gamma, z)$ be the orthogonal automorphy factor. The transformation law follows by differentiating

$$f_j(\gamma z) = \chi_j(\gamma) j(\gamma, z)^{k_j} f_j(z).$$

In each derivative row, the terms involving derivatives of $j(\gamma, z)$ are multiples of the first row and are removed by elementary row operations. Taking determinants, and using the standard tangent-Jacobian identity for a type-IV tube domain, gives the factor $j(\gamma, z)^{k_0 + \cdots + k_n + n}$, the product $\chi_0 \cdots \chi_n$, and the determinant character. This proves the stated weight and character. A reflection reverses the normal direction to its fixed hyperplane, and therefore the Jacobian is odd in a local normal coordinate. The final assertions are the standard modular-Jacobian reflection criterion; see [69, 67]. $\square$

C.2. The Weierstrass equation. Consider the family of $U \oplus E_8(-1)$ -polarized K3 surfaces. The very general member admits, up to automorphism, a unique Jacobian elliptic fibration with singular fibers $\text{II}^* + 14\text{I}_1$ [30, 10]. A Weierstrass model over $\mathbb{P}_{(t)}^1$ can be normalized as

$$(155) \quad y^2 = x^3 + f(t)x + g(t)$$

with

$$(156) \quad \begin{aligned} f(t) &= C_4 t^4 + C_{10} t^3 + C_{16} t^2 + C_{22} t + C_{28}, \\ g(t) &= t^7 + C_{12} t^5 + C_{18} t^4 + C_{24} t^3 + C_{30} t^2 + C_{36} t + C_{42}. \end{aligned}$$

This family was investigated by Hashimoto and Ueda in [30]. The coefficients C_n are assigned weight n , and t is assigned weight 6. With these conventions every monomial in f has weight 28, while every monomial in g has weight 42. Consequently, the discriminant polynomial

$$\Delta(t) := 4f(t)^3 + 27g(t)^2$$

is weighted homogeneous of total weight 84. Since $\Delta(t)$ has degree 14 in t , we may write

$$\Delta(t) = 27 \prod_{i=1}^{14} (t - r_i).$$

Under the scaling $C_n \mapsto \lambda^n C_n$, one has $t \mapsto \lambda^6 t$, $r_i \mapsto \lambda^6 r_i$. Therefore,

$$\text{Disc}_t(\Delta) = 27^{26} \prod_{i < j} (r_i - r_j)^2,$$

has weight 1092. In turn, the resultant $\text{Res}_t(f, g)$ has weight 196, as seen from the Sylvester determinant. Geometrically, the divisor defined by $\text{Disc}_t(\Delta)$ has two irreducible components: one corresponds to the collision of two nodal fibers, producing an additional A_1 -singularity, and the other to the simultaneous vanishing of f and g , producing a II-fiber. Thus one writes

$$\text{Disc}_t(\Delta) = \text{Res}_t(f, g)^3 \Psi_{504},$$

where Ψ_{504} is the equation of the divisor parameterizing surfaces acquiring an additional A_1 -singularity; its weight is $1092 - 3 \cdot 196 = 504$ [30, Section 3].

Hashimoto and Ueda proved in [30] that the scalar-valued ring of orthogonal modular forms for $\Gamma := \text{O}^+(U^{\oplus 2} \oplus E_8(-1))$ is

$$(157) \quad M_*(\Gamma) = \mathbb{C}[C_4, C_{10}, C_{12}, C_{16}, C_{18}, C_{22}, C_{24}, C_{28}, C_{30}, C_{36}, C_{42}],$$

and that the full ring with characters is obtained by adjoining a determinant-character generator Φ_{252} satisfying one relation of weight 504; namely $\Phi_{252}^2 = \Psi_{504}$; see also [20, 30, 72]. Applying Proposition C.1 to the eleven scalar generators, we obtain the following.

Theorem C.2. *The modular Jacobian $J = J(C_4, \dots, C_{42})$ of the Hashimoto–Ueda generators has weight 252 and determinant character. There is a constant $\kappa \in \mathbb{C}^\times$ such that $J = \kappa \Phi_{252}$. The scalar equation of the reflective branch divisor is $\Psi_{504} = \Phi_{252}^2 = 0$.*

Proof. The weight and character follow from Proposition C.1. The forms in (157) are algebraically independent, so their modular Jacobian is nonzero. Hashimoto–Ueda’s description of the full graded ring shows that the determinant-character module in weight 252 is one-dimensional and generated by Φ_{252} . Hence $J = \kappa \Phi_{252}$ for some $\kappa \neq 0$, and squaring gives the stated branch equation. $\square$

APPENDIX D. RELATION WITH COEFFICIENTS IN VINBERG’S SCROLL MODEL

Expanding (61) gives the following explicit polynomial change of parameters from Vinberg’s coefficients to those of (62):

$$(158) \quad \begin{aligned} F_4 &= -\frac{3}{2}a_2, & F_6 &= -\frac{3}{2}f_2, \\ G_8 &= f_1 + \frac{3}{4}a_2^2, & G_{10} &= a_1 + \frac{3}{2}a_2f_2, \quad G_{12} = a + \frac{3}{4}f_2^2, \\ H_8 &= b_2 - \frac{1}{4}a_2^2, & H_{10} &= g_2 - \frac{1}{2}a_2f_2, \\ H_{12} &= h - \frac{1}{2}a_2f_1 - \frac{1}{4}f_2^2 - \frac{1}{8}a_2^3, \\ H_{14} &= g_1 - \frac{1}{2}(a_2a_1 + f_2f_1) - \frac{3}{8}a_2^2f_2, \\ H_{16} &= b_1 - \frac{1}{2}(a_2a + f_2a_1) - \frac{3}{8}a_2f_2^2, \\ H_{18} &= b - \frac{1}{2}f_2a - \frac{1}{8}f_2^3. \end{aligned}$$

The change of parameters is triangular. Its inverse is

$$\begin{aligned}
(159) \quad & a_2 = -\frac{2}{3}F_4, & f_2 = -\frac{2}{3}F_6, \\
& f_1 = G_8 - \frac{1}{3}F_4^2, & a_1 = G_{10} - \frac{2}{3}F_4F_6, \quad a = G_{12} - \frac{1}{3}F_6^2, \\
& b_2 = H_8 + \frac{1}{9}F_4^2, & g_2 = H_{10} + \frac{2}{9}F_4F_6, \\
& h = H_{12} - \frac{1}{3}F_4G_8 + \frac{1}{9}F_6^2 + \frac{2}{27}F_4^3, \\
& g_1 = H_{14} - \frac{1}{3}(F_6G_8 + F_4G_{10}) + \frac{2}{9}F_4^2F_6, \\
& b_1 = H_{16} - \frac{1}{3}(F_6G_{10} + F_4G_{12}) + \frac{2}{9}F_4F_6^2, \\
& b = H_{18} - \frac{1}{3}F_6G_{12} + \frac{2}{27}F_6^3.
\end{aligned}$$

APPENDIX E. COEFFICIENT DICTIONARY FOR THE EXCEPTIONAL E_7 MODEL

We expand Equations (102) and (103) in the coefficients of the A_7 model. We obtain:

$$\begin{aligned}
(160) \quad & P_4 = f_4 - 4g_4, & P_6 = f_6 + 2g_6, \\
& P_{10} = f_4g_6 - 4f_6g_4 - 4h_5^2 + g_{10}, \\
& P_{12} = f_4^2g_4 - \frac{1}{3}f_4g_8 + 2f_6g_6 - \frac{8}{3}g_4g_8 + g_6^2 + 4h_5h_7 + g_{12}, \\
& P_{14} = h_7^2 - 4h_5h_9, \\
& P_{16} = 2f_4h_5h_7 - 4f_6h_5^2 - 4g_4g_{12} + g_6g_{10} - \frac{1}{3}g_8^2 + 2h_7h_9, \\
& P_{18} = f_4^2h_5^2 + 2f_4g_4g_{10} - \frac{1}{3}f_4g_6g_8 - 2f_4h_5h_9 - \frac{8}{3}f_6g_4g_8 + f_6g_6^2 \\
& \quad + 4f_6h_5h_7 - 4g_4h_7^2 + 4g_6h_5h_7 - \frac{8}{3}g_8h_5^2 + 2g_6g_{12} - \frac{1}{3}g_8g_{10} + h_9^2, \\
(161) \quad & P_{22} = -4g_4h_9^2 + 2g_6h_7h_9 - \frac{4}{3}g_8h_5h_9 - \frac{2}{3}g_8h_7^2 + 2g_{10}h_5h_7 - 4g_{12}h_5^2.
\end{aligned}$$

The two highest-weight coefficients are

$$\begin{aligned}
(162) \quad & P_{24} = 4f_4g_4h_7h_9 - 2f_4g_6h_5h_9 - \frac{2}{3}f_4g_8h_5h_7 + 2f_4g_{10}h_5^2 - 4f_6g_4h_7^2 \\
& \quad + 4f_6g_6h_5h_7 - \frac{8}{3}f_6g_8h_5^2 - \frac{8}{3}g_4g_8g_{12} + g_4g_{10}^2 + g_6^2g_{12} \\
& \quad - \frac{1}{3}g_6g_8g_{10} + 2g_6h_9^2 + \frac{2}{27}g_8^3 - \frac{2}{3}g_8h_7h_9 - 2g_{10}h_5h_9 + 4g_{12}h_5h_7, \\
& P_{30} = -\frac{8}{3}g_4g_8h_9^2 + 4g_4g_{10}h_7h_9 - 4g_4g_{12}h_7^2 + g_6^2h_9^2 - \frac{2}{3}g_6g_8h_7h_9
\end{aligned}$$

$$\begin{aligned}
& -2g_6g_{10}h_5h_9 + 4g_6g_{12}h_5h_7 + \frac{8}{9}g_8^2h_5h_9 + \frac{1}{9}g_8^2h_7^2 \\
(163) \quad & -\frac{2}{3}g_8g_{10}h_5h_7 - \frac{8}{3}g_8g_{12}h_5^2 + g_{10}^2h_5^2.
\end{aligned}$$

These identities provide a direct coefficient-level check of the weight decomposition in Equation (104). Under the specialization (110), coefficients factor as in Equation (113).

APPENDIX F. SELECTED FORMULAS FOR THE COEFFICIENT MAP Θ

We give selected exact relations for the geometric coefficient map Θ in Proposition 4.5. Computing the coefficients C_k from the binary-quartic invariants yields

$$\begin{aligned}
C_{16} = & \frac{6}{49}F_6^2F_4 + \frac{96}{49}F_6^2H_4 - \frac{3}{7}F_6G_{10} - \frac{16}{7}F_6H_{10} - 7G_{12}H_4 + H_{16} \\
(164) \quad & -\frac{1}{3}F_4^2H_4^2 - \frac{1}{3}F_4H_4G_8 + H_4^2G_8 + F_4H_{12} + 3H_4H_{12} - \frac{1}{3}G_8^2,
\end{aligned}$$

$$\begin{aligned}
C_{22} = & -\frac{4}{3}F_4H_4H_{14} + \frac{8}{7}F_6H_4H_{12} - \frac{1}{3}F_4G_{10}H_4^2 \\
& + \frac{1}{3}G_8G_{10}H_4 + \frac{2}{3}F_4F_6H_4^3 + \frac{22}{21}F_6G_8H_4^2 \\
& - G_{10}H_4^3 + G_{10}H_{12} - 2H_4^2H_{14} - \frac{2}{3}G_8H_{14} \\
& + \frac{3}{49}F_6^2G_{10} - \frac{4}{343}F_6^3F_4 + 2F_6G_{12}H_4 \\
& + \frac{2}{21}F_6F_4^2H_4^2 - \frac{2}{7}F_6F_4H_{12} - \frac{106}{343}F_6^3H_4 \\
& + \frac{44}{49}F_6^2H_{10} - \frac{2}{7}F_6H_{16} + \frac{2}{21}F_6G_8^2 \\
(165) \quad & - 4G_{12}H_{10} - 8H_{18}H_4 + \frac{2}{21}F_6F_4H_4G_8.
\end{aligned}$$

$$\begin{aligned}
C_{28} = & -\frac{1}{7}F_6G_{10}H_{12} + \frac{2}{21}F_6G_8H_{14} + F_4G_{12}H_4^3 \\
& + F_4H_4^2H_{16} + G_8G_{12}H_4^2 + G_8H_4H_{16} \\
& + G_{12}H_4H_{12} + G_{12}H_4^4 + H_4^3H_{16} + H_{12}H_{16} \\
& - \frac{1}{3}H_{14}^2 + \frac{1}{49}F_6^2F_4H_{12} - \frac{11}{49}F_6^2H_4H_{12} \\
& + \frac{4}{7}F_6G_{12}H_{10} - \frac{11}{21}F_6G_{10}H_4^3 - \frac{8}{21}F_6H_4^2H_{14} \\
& - \frac{2}{3}G_{10}H_4H_{14} - \frac{1}{7}F_6^2G_{12}H_4 - \frac{1}{147}F_6^2F_4^2H_4^2 \\
& - \frac{25}{147}F_6^2H_4^2G_8 - \frac{1}{3}G_{10}^2H_4^2 - \frac{1}{147}F_6^2F_4H_4G_8 \\
& + \frac{1}{21}F_6F_4G_{10}H_4^2 - \frac{1}{21}F_6G_8G_{10}H_4 + \frac{4}{21}F_6F_4H_4H_{14} \\
& \quad \quad \quad 46
\end{aligned}$$

$$\begin{aligned}
& + \frac{8}{7}F_6H_{18}H_4 - \frac{2}{21}F_4F_6^2H_4^3 - 4H_{18}H_{10} \\
& - \frac{1}{3}F_6^2H_4^4 + \frac{1}{2401}F_6^4F_4 - \frac{1}{343}F_6^3G_{10} \\
& + \frac{37}{2401}F_6^4H_4 + \frac{1}{49}F_6^2H_{16} - \frac{24}{343}F_6^3H_{10} \\
& - \frac{1}{147}F_6^2G_8^2.
\end{aligned}
\tag{166}$$

APPENDIX G. EISENSTEIN EXPRESSIONS FOR THE HASHIMOTO–UEDA GENERATORS

We give the exact relations for the generators used in Proposition 2.5. Throughout this appendix, $\mathcal{E}_k$ denotes the cusp-normalized orthogonal Eisenstein series of (19). These are not Sakai's series $\mathcal{E}_k^S$; the two normalizations differ by the Bernoulli factor in that equation. Direct computation gives

$$\begin{aligned}
C_4 &= -3\mathcal{E}_4, \\
C_{10} &= \frac{24}{7}\mathcal{E}_{10}, \\
(167) \quad C_{12} &= -\frac{4146}{875}\mathcal{E}_{12} + \frac{378}{125}\mathcal{E}_4^3, \\
C_{16} &= -\frac{22494123}{1990625}\mathcal{E}_{16} + \frac{3756276}{284375}\mathcal{E}_4\mathcal{E}_{12} - \frac{10557}{3125}\mathcal{E}_4^4, \\
C_{18} &= \frac{175468}{13475}\mathcal{E}_{18} - \frac{3132}{275}\mathcal{E}_4^2\mathcal{E}_{10}, \\
C_{22} &= \frac{245355851592}{6407025625}\mathcal{E}_{22} - \frac{528963264}{48173125}\mathcal{E}_{10}\mathcal{E}_{12} - \frac{469552368}{10058125}\mathcal{E}_4\mathcal{E}_{18} + \frac{10401912}{529375}\mathcal{E}_4^3\mathcal{E}_{10}.
\end{aligned}$$

In weights 24, 28, and 30, one has

$$\begin{aligned}
(168) \quad C_{24} &= -\frac{247630621761606}{5887382125625}\mathcal{E}_{24} + \frac{88284326976}{10221859375}\mathcal{E}_{12}^2 + \frac{6061824}{468391}\mathcal{E}_4\mathcal{E}_{10}^2 \\
& + \frac{119646240237}{2673409375}\mathcal{E}_4^2\mathcal{E}_{16} - \frac{3193630632}{112328125}\mathcal{E}_4^3\mathcal{E}_{12} + \frac{4270941}{1234375}\mathcal{E}_4^6, \\
(169) \quad C_{28} &= -\frac{28869530234706049572}{212555734025009375}\mathcal{E}_{28} + \frac{47840342617727064}{1650995262109375}\mathcal{E}_{12}\mathcal{E}_{16} + \frac{3191204229888}{84553024325}\mathcal{E}_{10}\mathcal{E}_{18} \\
& + \frac{4554670026061219158}{25250981936805625}\mathcal{E}_4\mathcal{E}_{24} - \frac{21493441243002816}{1096038871484375}\mathcal{E}_4\mathcal{E}_{12}^2 - \frac{258000273792}{7174746425}\mathcal{E}_4^2\mathcal{E}_{10}^2 \\
& - \frac{134448206960657241}{1433281601171875}\mathcal{E}_4^3\mathcal{E}_{16} + \frac{70065352514484}{1720626171875}\mathcal{E}_4^4\mathcal{E}_{12} - \frac{2446356636099}{926491015625}\mathcal{E}_4^7, \\
(170) \quad C_{30} &= \frac{27209419519503579144}{186144949274659375}\mathcal{E}_{30} - \frac{25369042718016}{562070884375}\mathcal{E}_{12}\mathcal{E}_{18} - \frac{392398848}{88110869}\mathcal{E}_{10}^3 \\
& - \frac{1899223793136}{20116221125}\mathcal{E}_4\mathcal{E}_{10}\mathcal{E}_{16} - \frac{396004344469488}{2187752826875}\mathcal{E}_4^2\mathcal{E}_{22} + \frac{4032113824512}{46487065625}\mathcal{E}_4^2\mathcal{E}_{10}\mathcal{E}_{12} \\
& + \frac{8765776266336}{67942634375}\mathcal{E}_4^3\mathcal{E}_{18} - \frac{18776847816}{510846875}\mathcal{E}_4^5\mathcal{E}_{10}.
\end{aligned}$$

The two longest identities are

$$\begin{aligned}
C_{36} = & -\frac{10966101442664558639832373507186}{20664063689028667667525390625}\mathcal{E}_{36} + \frac{711906653618754304}{11152800682340625}\mathcal{E}_{18}^2 \\
& + \frac{9952374401195974480016}{74473435858305859375}\mathcal{E}_{12}\mathcal{E}_{24} - \frac{16025286253876891776}{2198153360634765625}\mathcal{E}_{12}^3 \\
& + \frac{173837469469737984}{3793119345640625}\mathcal{E}_{10}^2\mathcal{E}_{16} + \frac{17805779546323923071901}{108248886284423828125}\mathcal{E}_4\mathcal{E}_{16}^2 \\
& + \frac{386071447810308586944}{1072560276146734375}\mathcal{E}_4\mathcal{E}_{10}\mathcal{E}_{22} - \frac{442884732896131584}{5539866741784375}\mathcal{E}_4\mathcal{E}_{10}^2\mathcal{E}_{12} \\
& + \frac{316115129874677291208708972}{416187922859075945703125}\mathcal{E}_4^2\mathcal{E}_{28} - \frac{90605903434364369268733632}{308219146123005126953125}\mathcal{E}_4^2\mathcal{E}_{12}\mathcal{E}_{16} \\
& - \frac{830577945310736931456}{2197903021688171875}\mathcal{E}_4^2\mathcal{E}_{10}\mathcal{E}_{18} - \frac{36320037840362329936000044}{59671226689413792578125}\mathcal{E}_4^3\mathcal{E}_{24} \\
& + \frac{271723776565975828624176}{2590076858176513671875}\mathcal{E}_4^3\mathcal{E}_{12}^2 + \frac{169076009521112340096}{1305521343709515625}\mathcal{E}_4^4\mathcal{E}_{10}^2 \\
& + \frac{751395218162920164090549}{3387023583769287109375}\mathcal{E}_4^5\mathcal{E}_{16} - \frac{135095208176303563806}{1498020160888671875}\mathcal{E}_4^6\mathcal{E}_{12} \\
(171) \quad & + \frac{1582982591047959024}{312773440185546875}\mathcal{E}_4^9,
\end{aligned}$$

and

$$\begin{aligned}
C_{42} = & \frac{937658283184177005411726687689439004}{472563301052781894820381313671875}\mathcal{E}_{42} \\
& - \frac{174377132156027221342821058336}{433495455925792705523203125}\mathcal{E}_{18}\mathcal{E}_{24} \\
& - \frac{524243665969361758958944518286208}{1258499497774101443340558203125}\mathcal{E}_{12}\mathcal{E}_{30} \\
& + \frac{509119589590898191681843968}{9684077944495913161328125}\mathcal{E}_{12}^2\mathcal{E}_{18} \\
& - \frac{255976422600718744575572808}{1690659567345019755859375}\mathcal{E}_{10}\mathcal{E}_{16}^2 \\
& - \frac{253979387411191905980928}{1522866217812128564375}\mathcal{E}_{10}^2\mathcal{E}_{22} \\
& + \frac{2150945991433083703296}{96081458005145738125}\mathcal{E}_{10}^3\mathcal{E}_{12} \\
& - \frac{579685483277498303643935181348}{478058855307500586259765625}\mathcal{E}_4\mathcal{E}_{16}\mathcal{E}_{22} \\
& - \frac{9342416050138099539846158310096}{6500132405485958107978484375}\mathcal{E}_4\mathcal{E}_{10}\mathcal{E}_{28} \\
& + \frac{9295217211099331001609272699968}{18480648441128142380439453125}\mathcal{E}_4\mathcal{E}_{10}\mathcal{E}_{12}\mathcal{E}_{16} \\
& + \frac{84336513837366609025425408}{246533341350932314188125}\mathcal{E}_4\mathcal{E}_{10}^2\mathcal{E}_{18} \\
& + \frac{10766383697512235671478541672}{18044266397177518544921875}\mathcal{E}_4^2\mathcal{E}_{16}\mathcal{E}_{18}
\end{aligned}$$

$$\begin{aligned}
& + \frac{411133600292986721437818995664}{775339151885274060068359375} \mathcal{E}_4^2 \mathcal{E}_{12} \mathcal{E}_{22} \\
& + \frac{725349075991254266821519329528}{931960907501616498240109375} \mathcal{E}_4^2 \mathcal{E}_{10} \mathcal{E}_{24} \\
& - \frac{633175334098004178493948224768}{3435226416115584113069921875} \mathcal{E}_4^2 \mathcal{E}_{10} \mathcal{E}_{12}^2 \\
& - \frac{814796153444743328014789939632}{446118219700142305331640625} \mathcal{E}_4^3 \mathcal{E}_{30} \\
& - \frac{1545660921032085339748204896}{29265070711388748564453125} \mathcal{E}_4^3 \mathcal{E}_{12} \mathcal{E}_{18} \\
& - \frac{123243972544099392334848}{2956623058517985250625} \mathcal{E}_4^3 \mathcal{E}_{10}^3 \\
& + \frac{73492865380951694535471199362576}{112305478988394096004208984375} \mathcal{E}_4^4 \mathcal{E}_{10} \mathcal{E}_{16} \\
& + \frac{1999656919995936407243390016}{1217172922896819560546875} \mathcal{E}_4^5 \mathcal{E}_{22} \\
& - \frac{5620222544686734151056072672}{12256409362478892939453125} \mathcal{E}_4^5 \mathcal{E}_{10} \mathcal{E}_{12} \\
& - \frac{303046910652213425086168068}{321594183641634599609375} \mathcal{E}_4^6 \mathcal{E}_{18} \\
& + \frac{275930723848426493011104564}{1481543988871074970703125} \mathcal{E}_4^8 \mathcal{E}_{10}.
\end{aligned}
\tag{172}$$

Each monomial in these equations has the indicated total weight. After the normalization change (19), we find that the identities agree term by term with Sakai's Appendix A.

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DEPARTMENT OF MATHEMATICS AND STATISTICS, UNIVERSITY OF MISSOURI - ST. LOUIS, ST. LOUIS,
MO 63121

Email address: `clinghera@umsl.edu`

DEPARTMENT OF MATHEMATICS & STATISTICS, UTAH STATE UNIVERSITY, LOGAN, UT 84322

Email address: `andreas.malmendier@usu.edu`

INSTITUTE FOR MATHEMATICS, HEIDELBERG UNIVERSITY, 69120 HEIDELBERG, GERMANY

Email address: `bwilliams@mathi.uni-heidelberg.de`